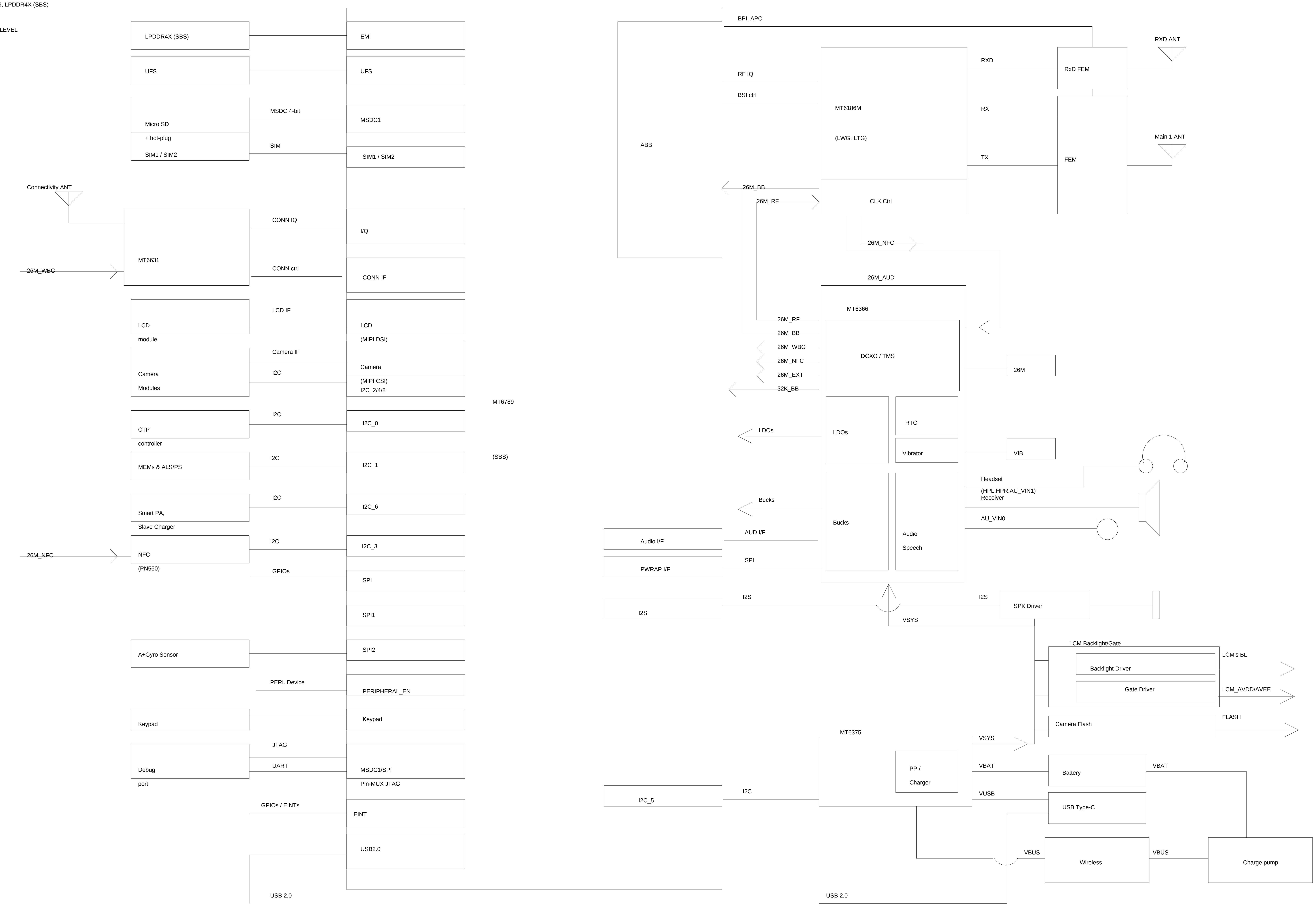
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

Pre-released for BOM Cost evaluation only

Project : MT6789, LPDDR4X (SBS)

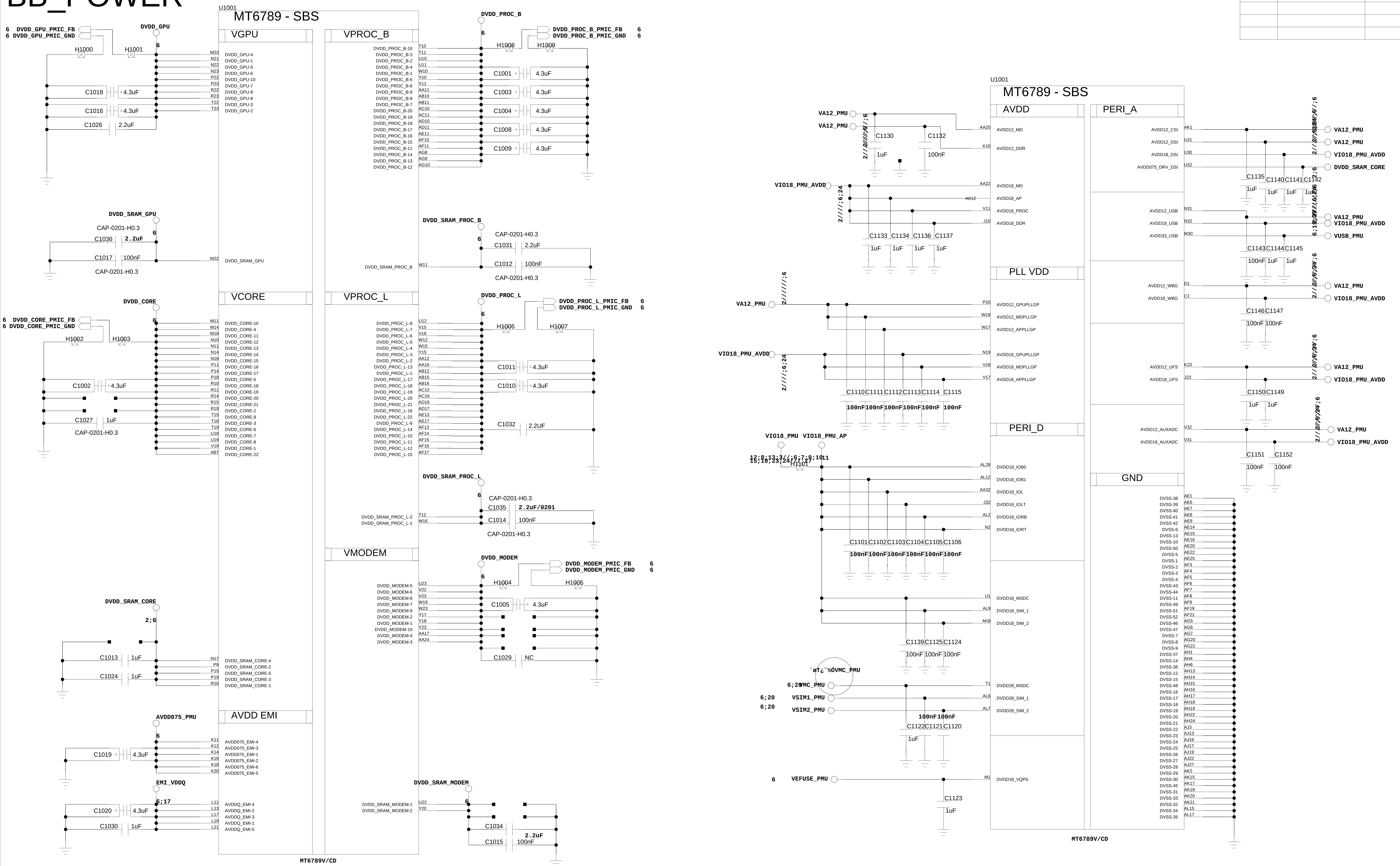
REF_SCH TOP LEVEL



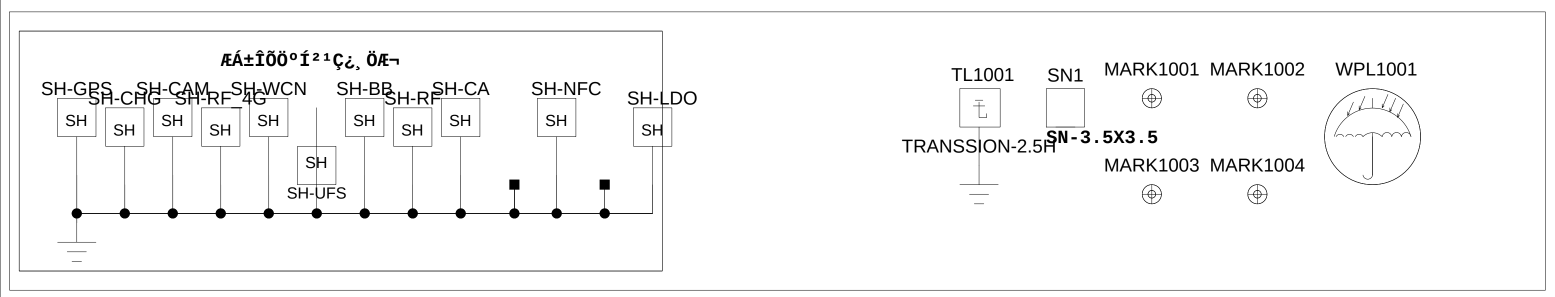
COMPANY: TRANSSION HOLDINGS				MODEL: **G99_STD_MAIN**	Modified Date: **02/05/2024:09:59**
DRAWN		DATED		TITLE: **00_BLOCK_DIAGRAM**	VERSION:
CHECKED		DATED	< >	Confidentiality CONFIDENTIAL	SHEET: **1** OF **27**

BB_POWER

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

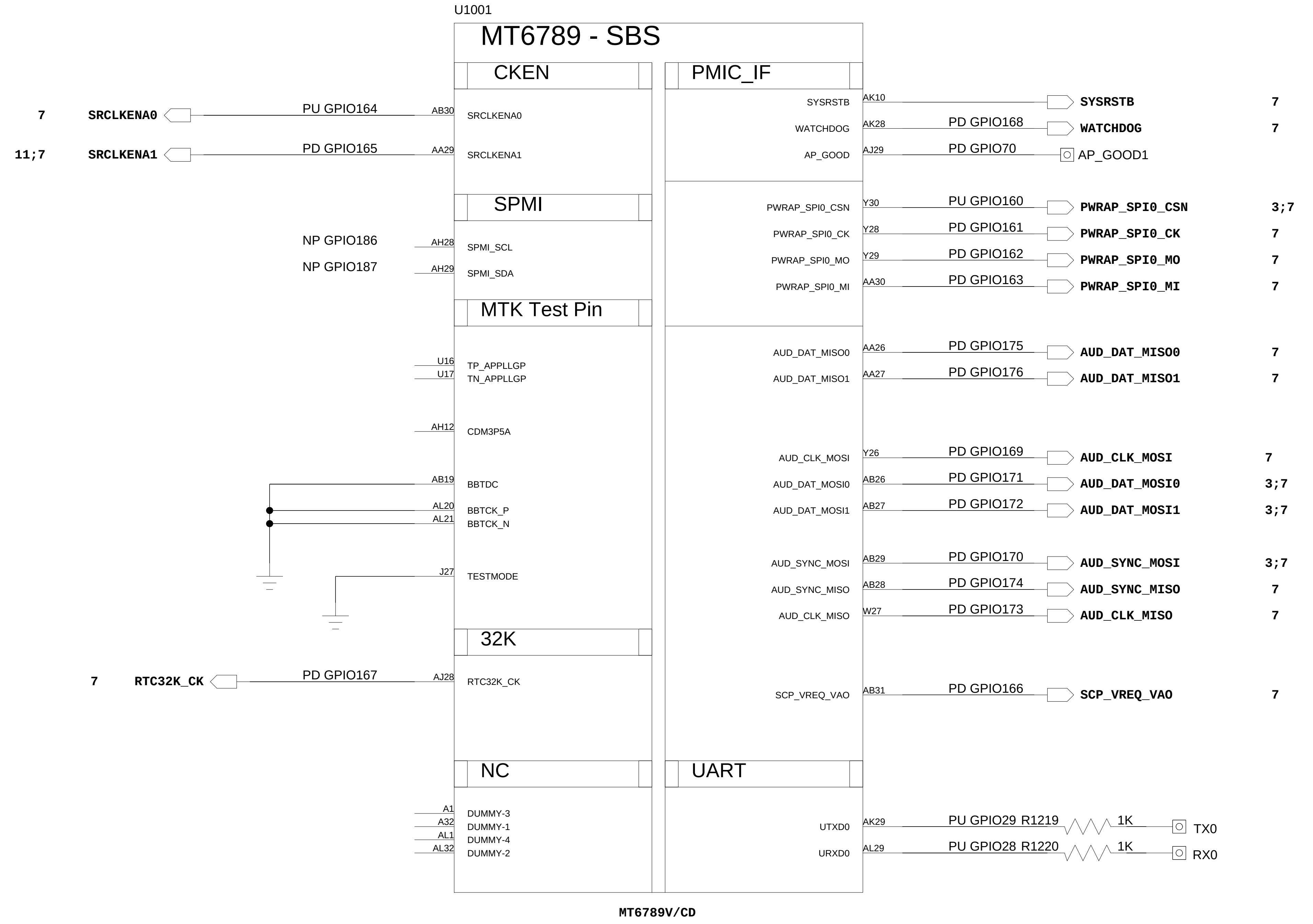


SHIELDING&LOGO&SN&MARK



COMPANY: TRANSSION HOLDINGS		MODEL: G99_STD_MAIN	Modified Date: 02/05/2024:09:57
DRAWN	DATED	TITLE: 10_BB_POWER_PDN	VERSION:
CHECKED	DATED < >	Confidentiality CONFIDENTIAL	SHEET: 2 OF 27

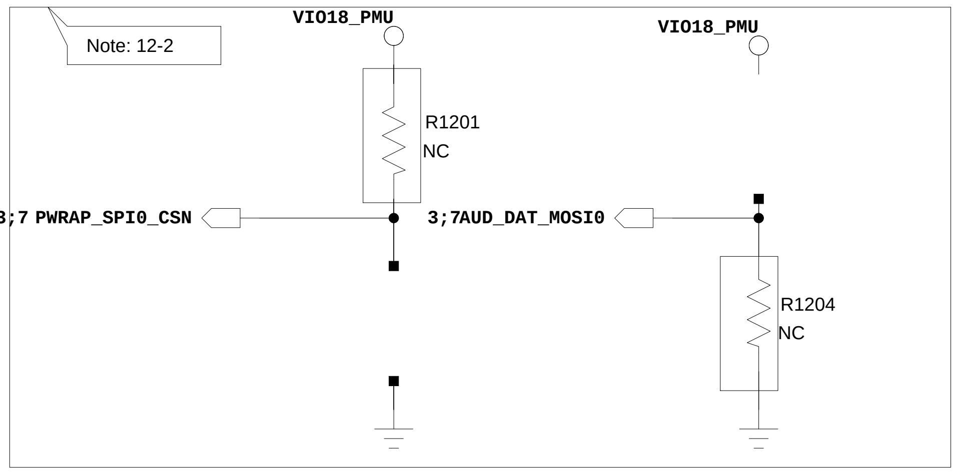
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



Schematic design notice of "12_BB_1" page:

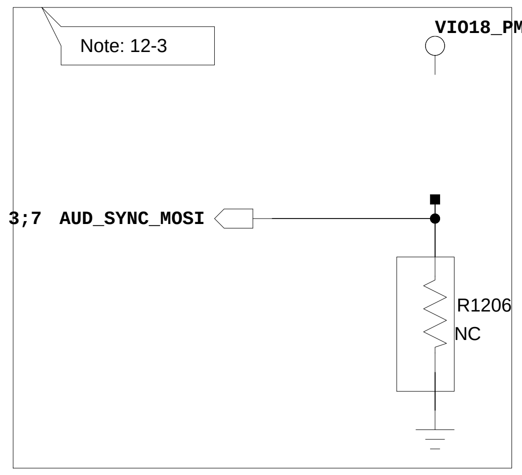
Note 12-2: "PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" pin features in trapping pin to enable JTAG.

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	AP JTAG Mode	IO JTAG Mode
H (Default)	L (Default)	N/A	N/A
H	H (by external PU)	SPI5_CSB, SPI5_CLK, SPI5_MO, SPI5_MI, EINT13	N/A
L (by external PD)	L	SPI5_CSB, SPI5_CLK, SPI5_MO, SPI5_MI, EINT13	SPI1_CSB, SPI1_CLK, SPI1_MO, SPI1_MI, EINT14
L (by external PD)	H (by external PU)	N/A	N/A



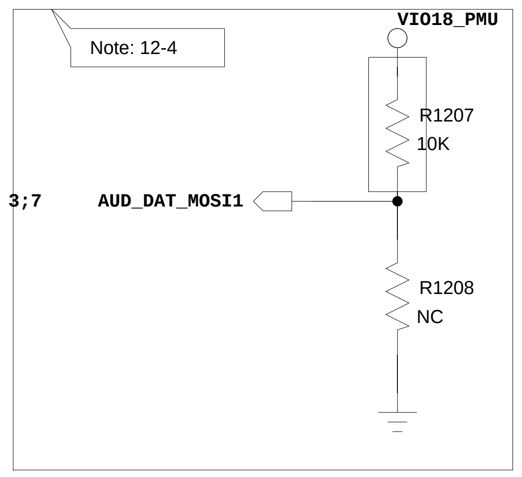
Note 12-3: "AUD_SYNC_MOSI" pin features in trapping pin to booting (UFS/eMMC).

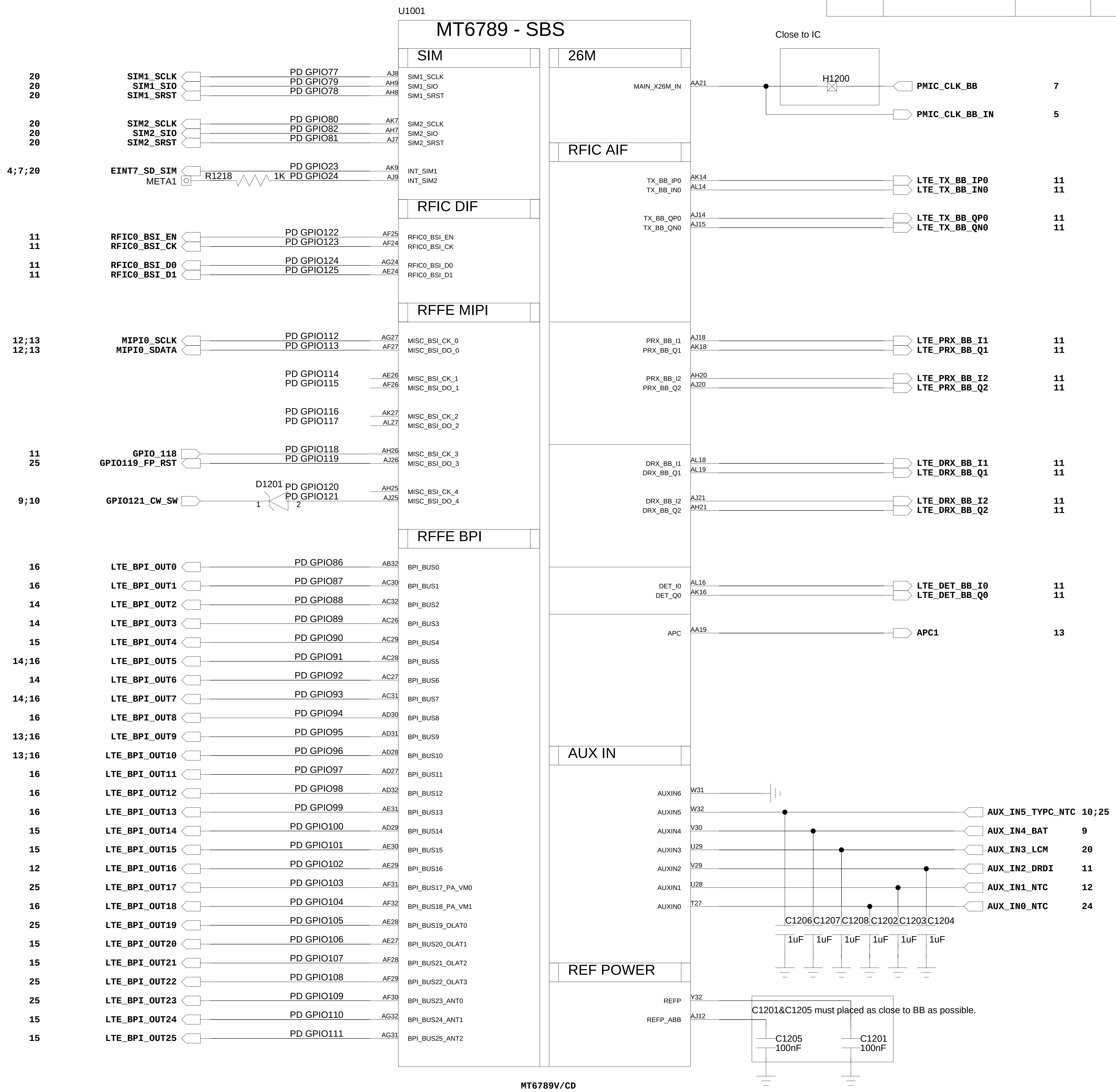
AUD_SYNC_MOSI	Storage Booting Mode
L (Default)	Only UFS boot
H (by external PU)	Only eMMC boot



Note 12-4 "AUD_DAT_MOSI1" pin features in trapping pin for SW co-load.

AUD_DAT_MOSI1	Mode
L (Default)	LP4x-MCP
H (by external PU)	LP4x-Discrete

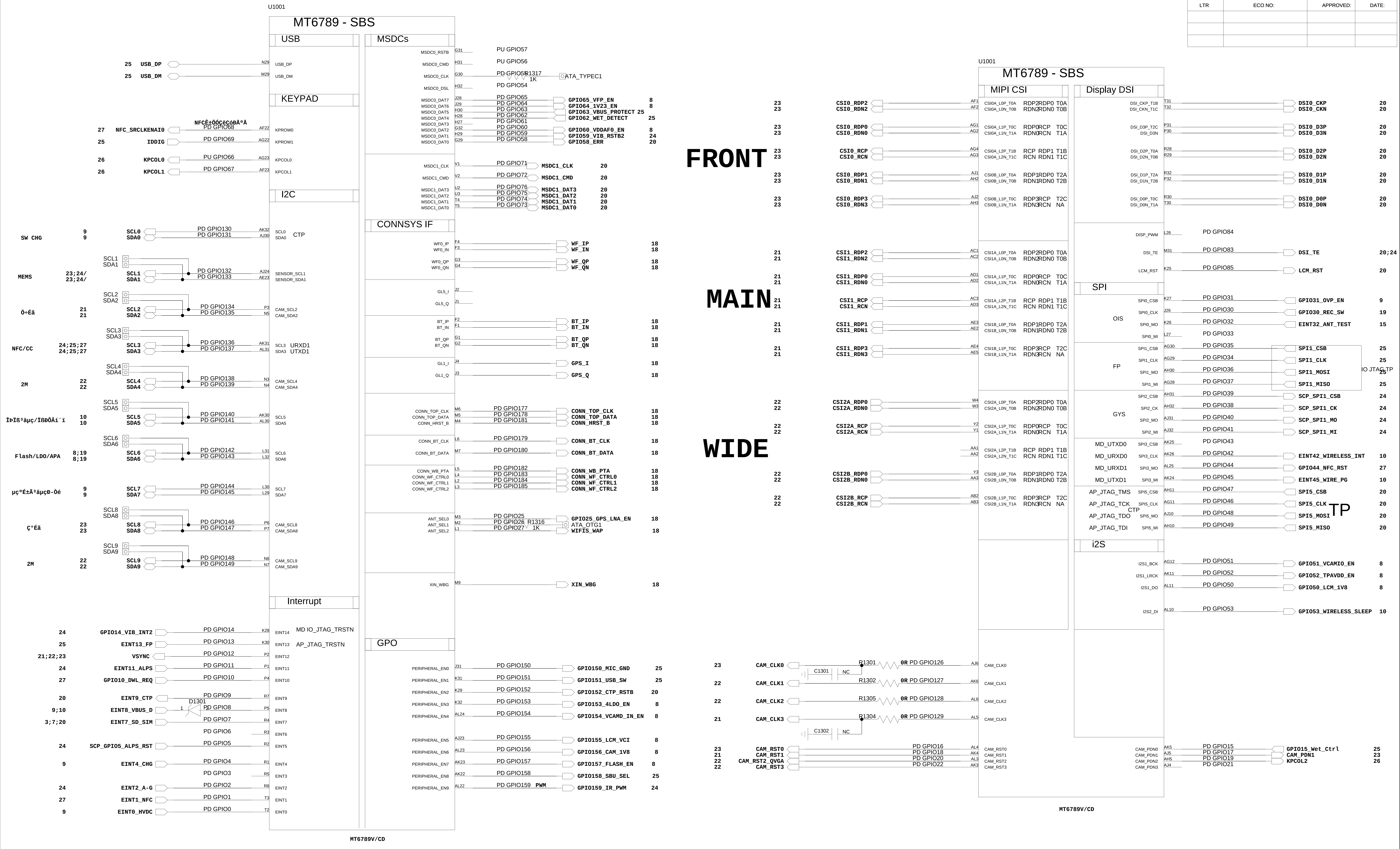




COMPANY: TRANSSION HOLDINGS		MODEL: G99_STD_MAIN	Modified Date: 24/04/2024:16:10
DRAWN	DATED	TITLE: 12_BB_I_PMIC_RF	VERSION:
CHECKED	DATED < >	Confidentiality CONFIDENTIAL	SHEET: 3 OF 27

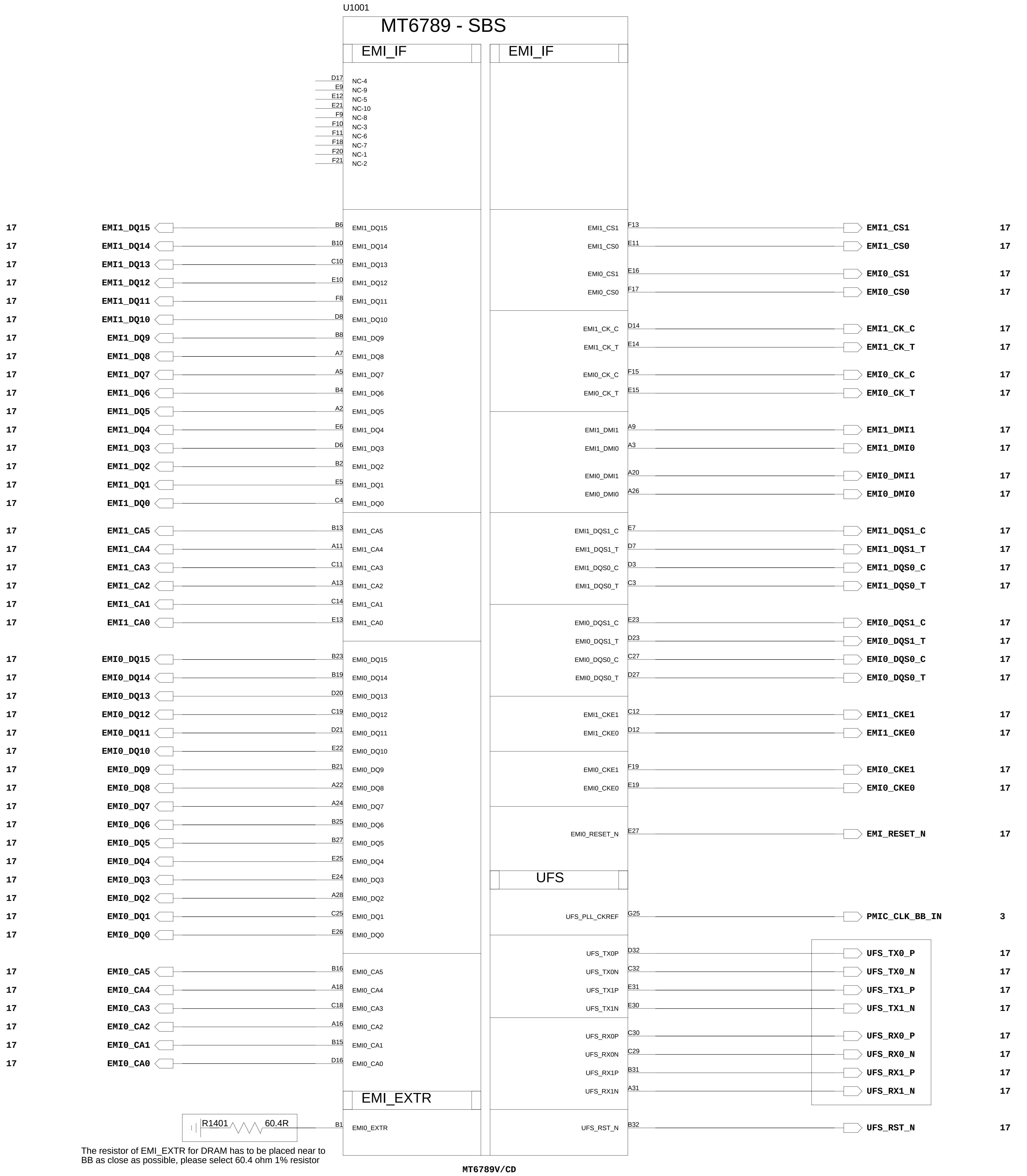
REVISION RECORD

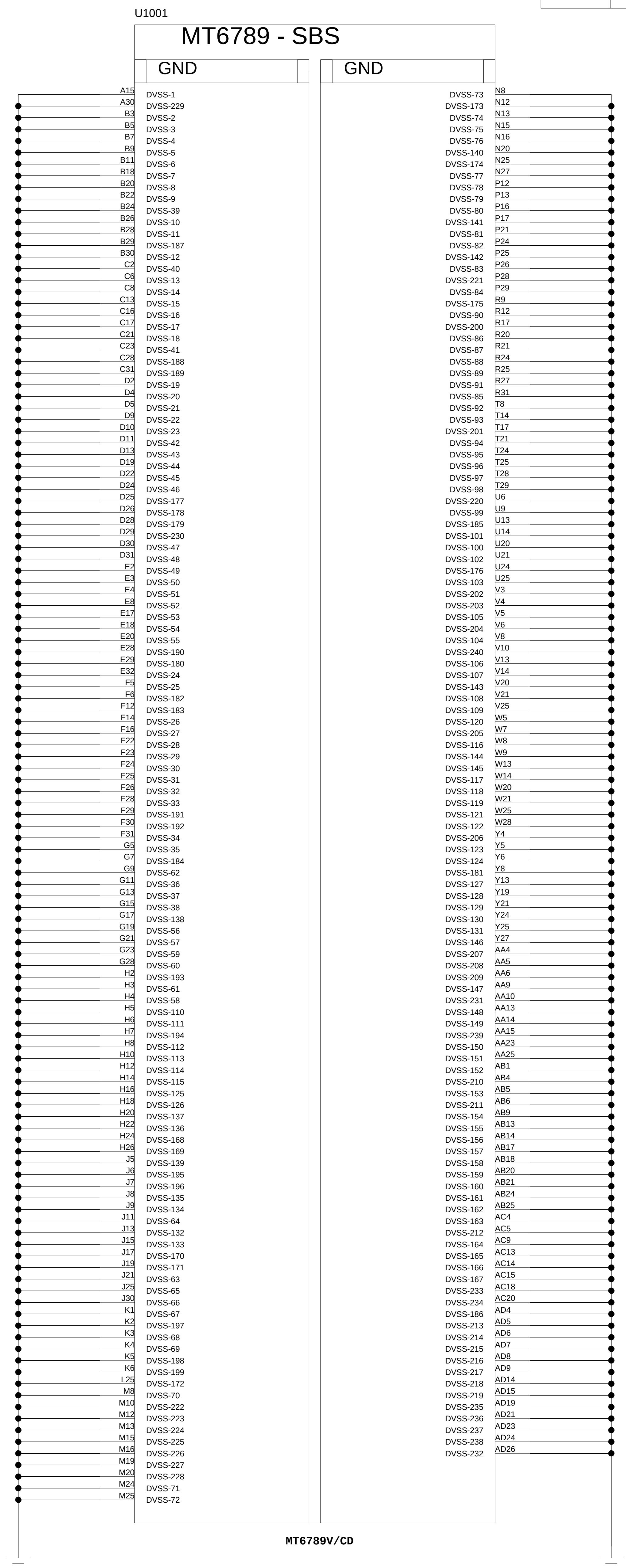
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN	Modified Date: 24/04/2024:21:14
DRAWN		DATED		TITLE: 13_BB_II_CONN_RF	VERSION:
CHECKED		DATED	< >	Confidentiality: CONFIDENTIAL	SHEET: 4 OF 27

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:





Schematic design notice of "14_BB_3_Interface" page:

Note 14-1: R4001 please select 60.4 ohm (1%) resistor

Note 14-2: Make sure TX/RX connection (T to in, R to out and P to t, N to c)

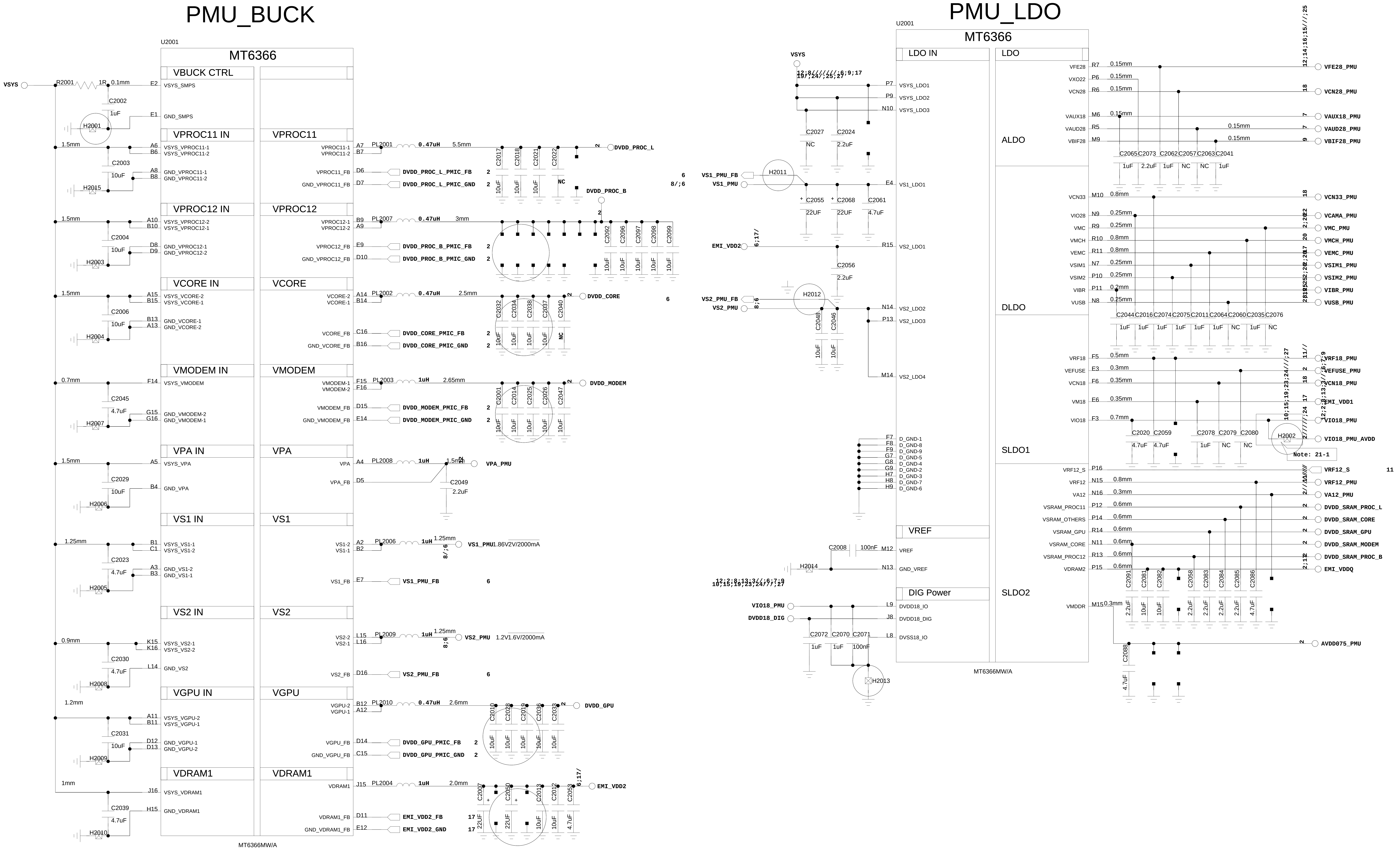
COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN	Modified Date: 17/04/2024:14:51
DRAWN		DATED		TITLE: 14_BB_III_DDR	VERSION:
CHECKED		DATED	< >	Confidentiality CONFIDENTIAL	SHEET: 5 OF 27

POWER_MT6366_I

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

PMU_BUCK

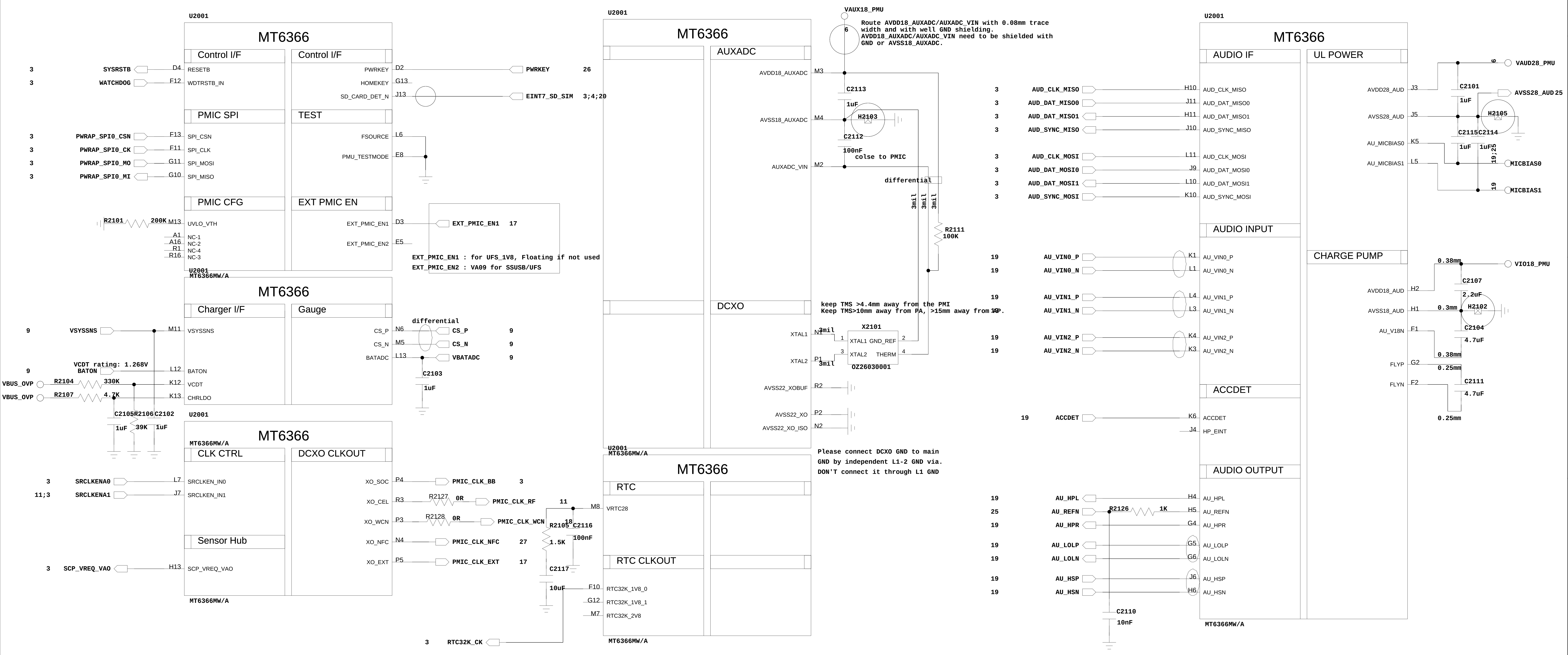
PMU_LDO



COMPANY: TRANSSION HOLDINGS		MODEL: G99_STD_MAIN	Modified Date:	05/06/2024:11:07
DRAWN	DATED	TITLE: 20_POWER_MT6366_BUCK_LDO		
CHECKED	DATED < >	Confidentiality CONFIDENTIAL	VERSION:	SHEET: 6 OF 27

POWER_MT6366_II

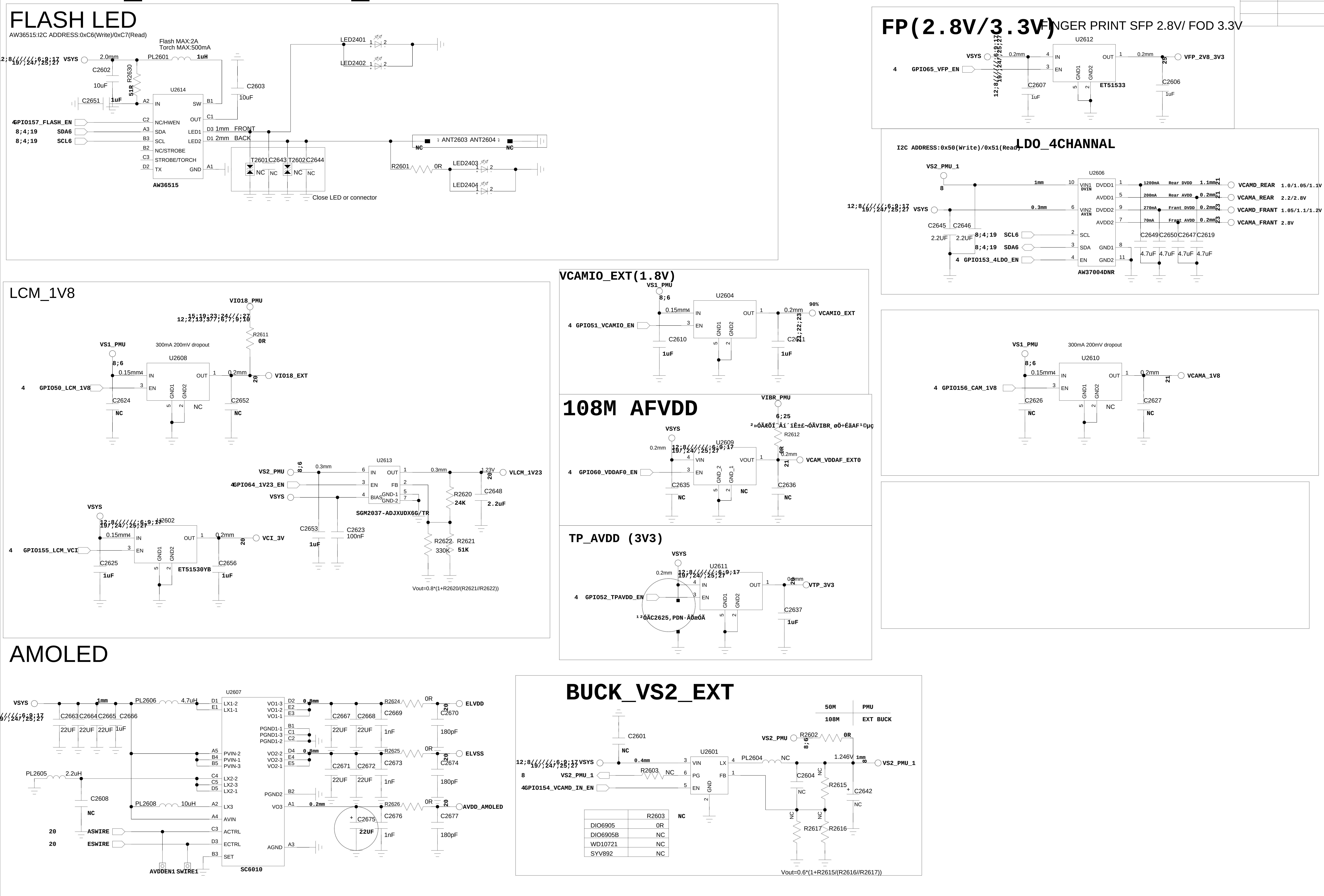
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS		MODEL: G99_STD_MAIN	Modified Date: 29/04/2024:19:36
DRAWN	DATED	TITLE: 21_POWER_MT6366_GENERAL_CLK	VERSION:
CHECKED	DATED < >	Confidentiality CONFIDENTIAL	SHEET: 7 OF 27

POWER_THIRD-PARTY_II

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



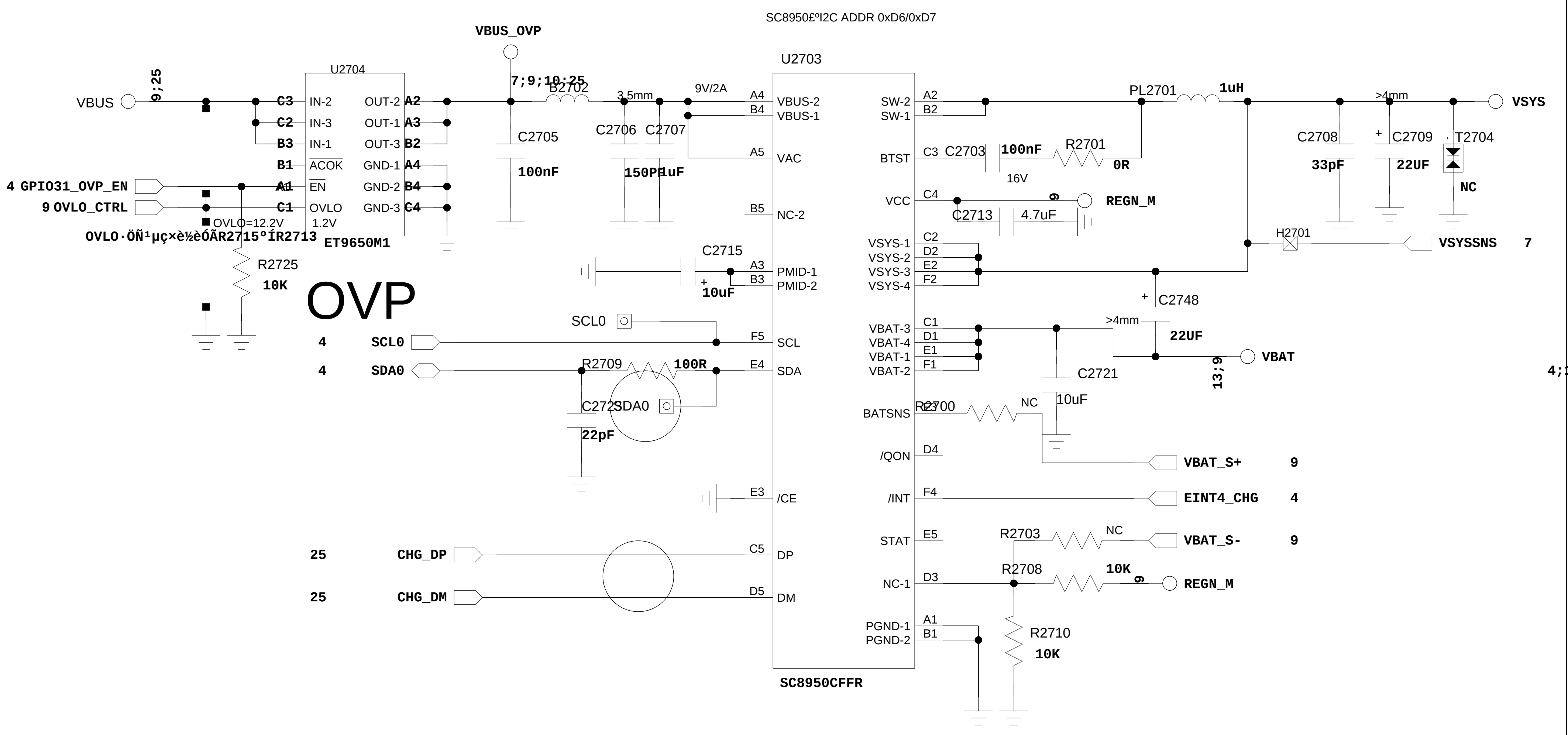
COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN	Modified Date: 30/04/2024:20:04
DRAWN		DATED		TITLE: 26_POWER_THIRD-PARTY-II	VERSION:
CHECKED		DATED	< >	Confidentiality CONFIDENTIAL	SHEET: 8 OF 27

POWER_THIRD-PARTY_II

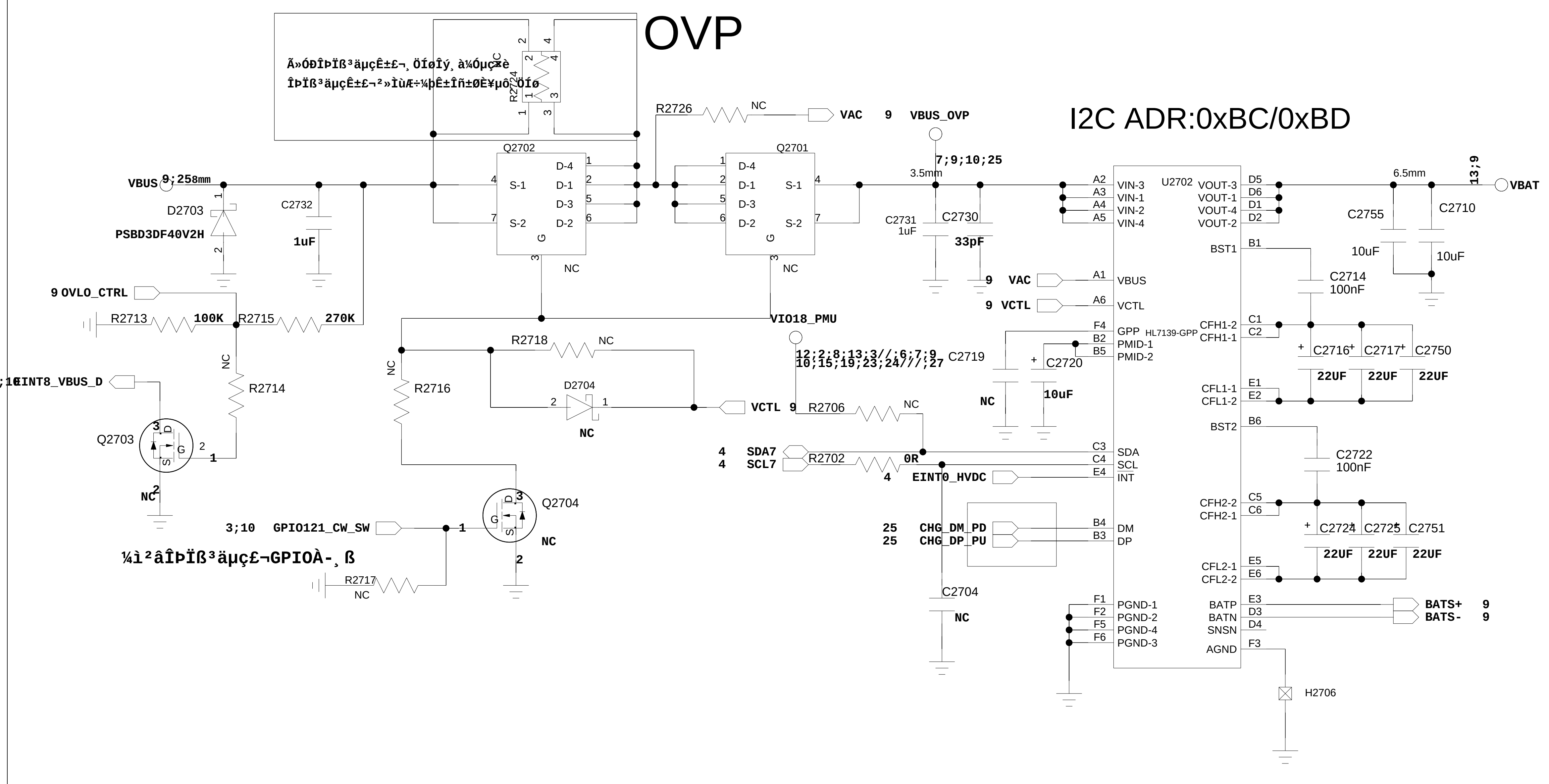
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

CHARGER_I

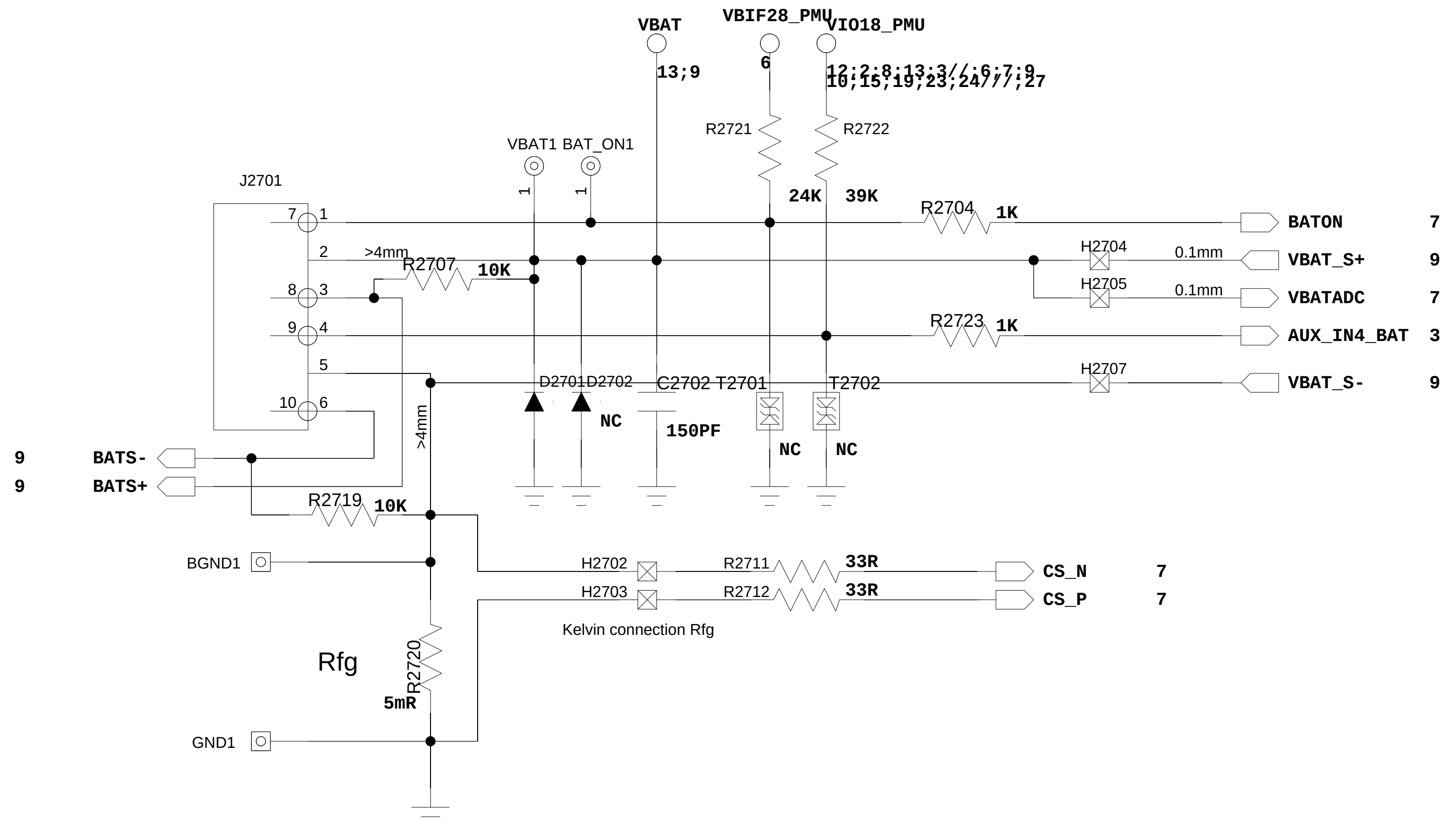
	R2715	R2713
ÎÞÏß³äµçMOS	270K	100K
18/33W OVP IC	330K	36K



OVP



BATTERY CONNECTOR



COMPANY: TRANSSION HOLDINGS			MODEL: G99_STD_MAIN	Modified Date:	25/04/2024:16:41
DRAWN		DATED	TITLE: 27_POWER_THIRD-PARTY_III		
CHECKED		DATED < >	Confidentiality CONFIDENTIAL	VERSION:	SHEET: 9 OF 27

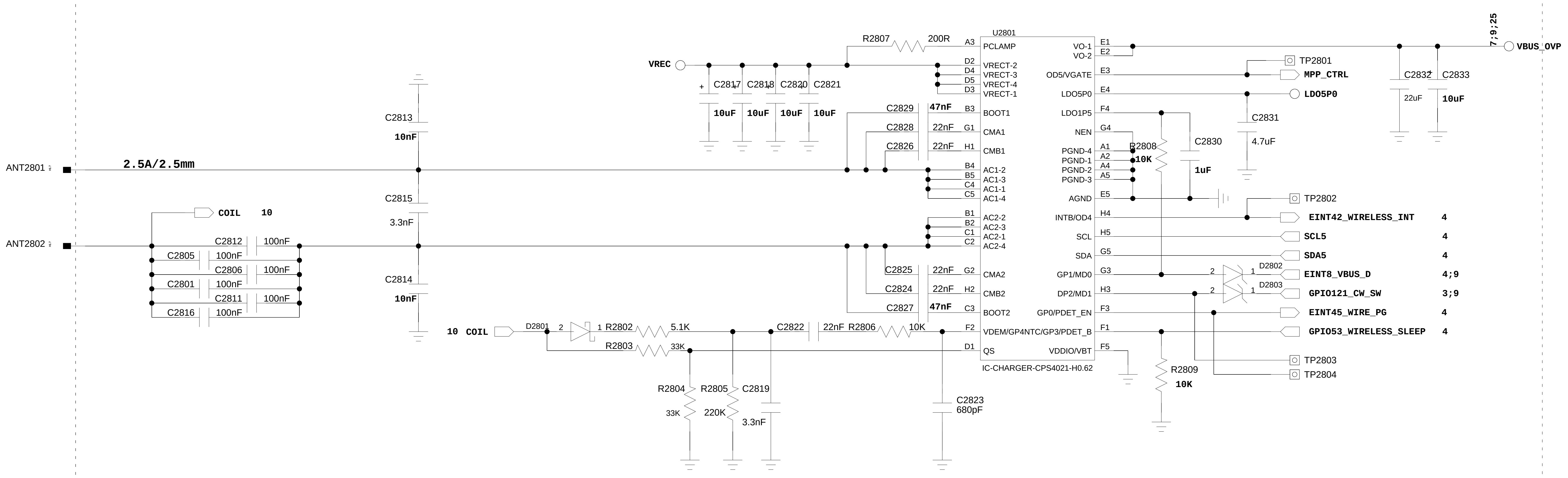
POWER_THIRD-PARTY-IV

WIRELESS CHG

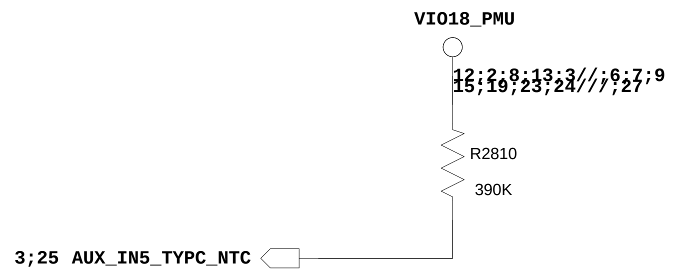
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

NC

CPS4021:I2C5 ADDRESS 0x60/0x61



Thermistor to sense
SUB TYPEC temperature



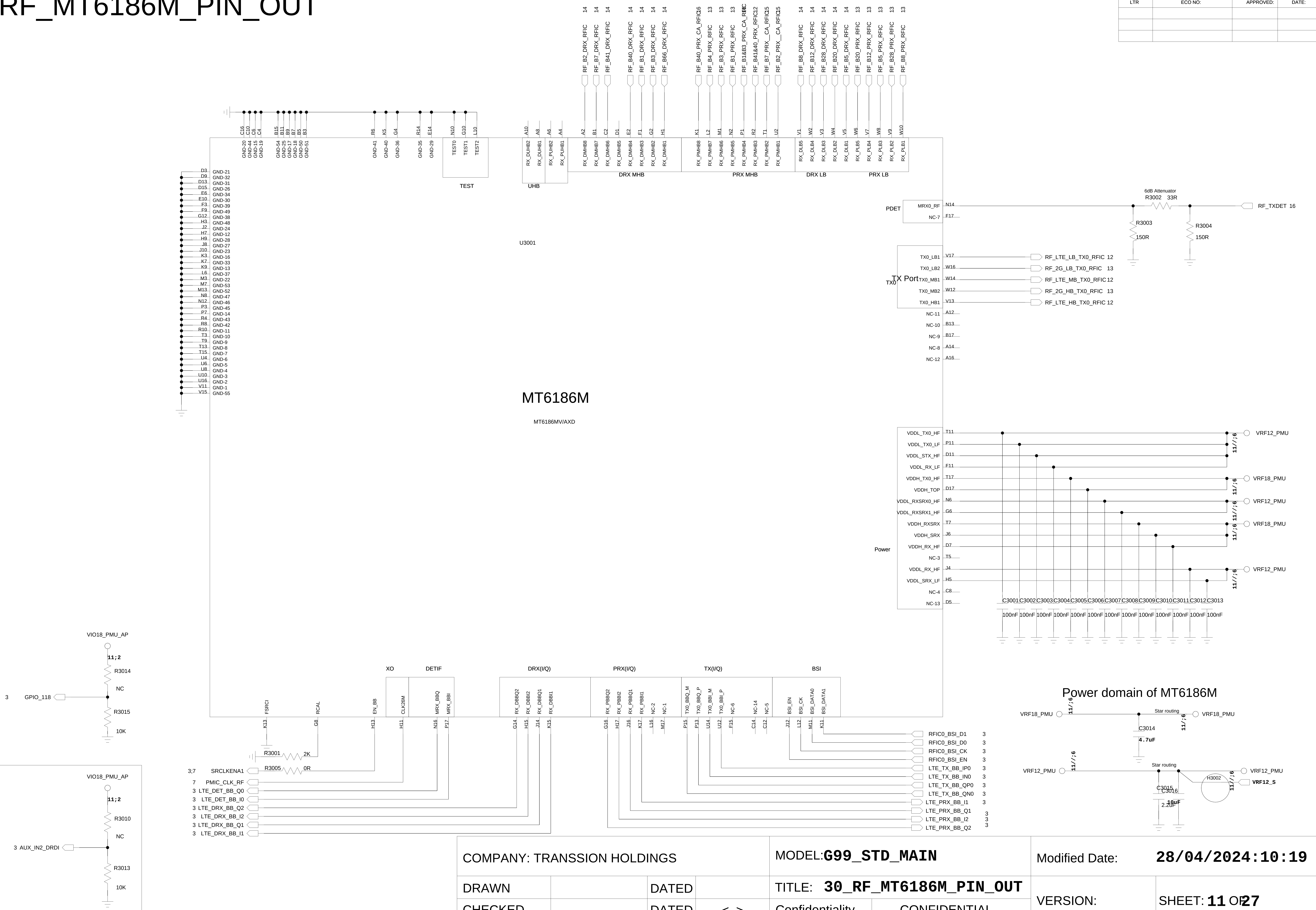
	R2810	SUB°åNTC
Type-c	390K	100K NTC

Thermistor to sense
Top Holding area temperature

COMPANY: TRANSSION HOLDINGS				MODEL: **G99_STD_MAIN**	Modified Date: **17/04/2024:14:51**
DRAWN		DATED		TITLE: **28_POWER_THIRD-PARTY-IV**	VERSION:
CHECKED		DATED	< >	Confidentiality CONFIDENTIAL	SHEET: **10** OF **27**

RF_MT6186M_PIN_OUT

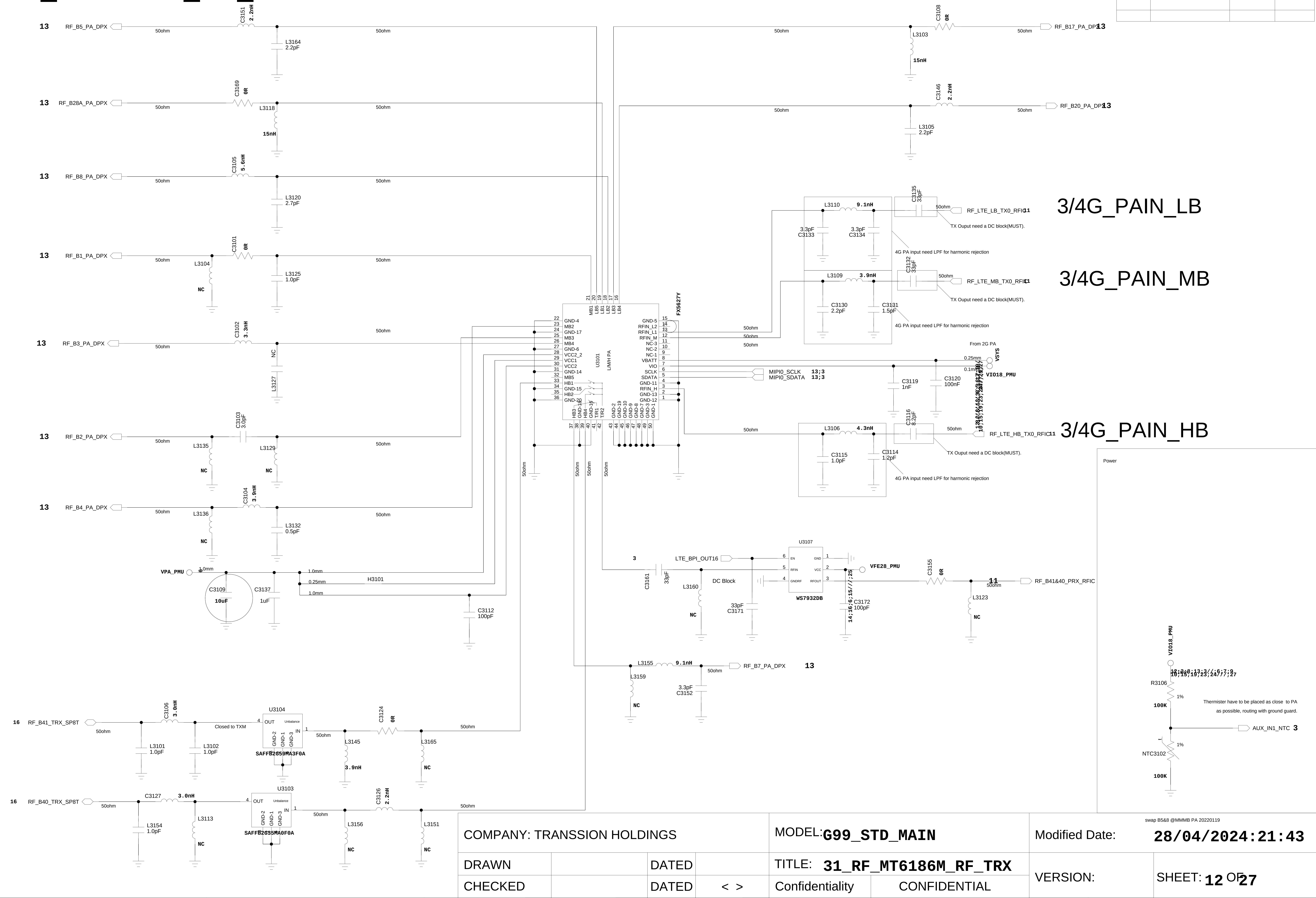
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS		MODEL: **G99_STD_MAIN**	Modified Date: **28/04/2024:10:19**
DRAWN	DATED	TITLE: **30_RF_MT6186M_PIN_OUT**	VERSION:
CHECKED	DATED < >	Confidentiality CONFIDENTIAL	SHEET: **11** OF **27**

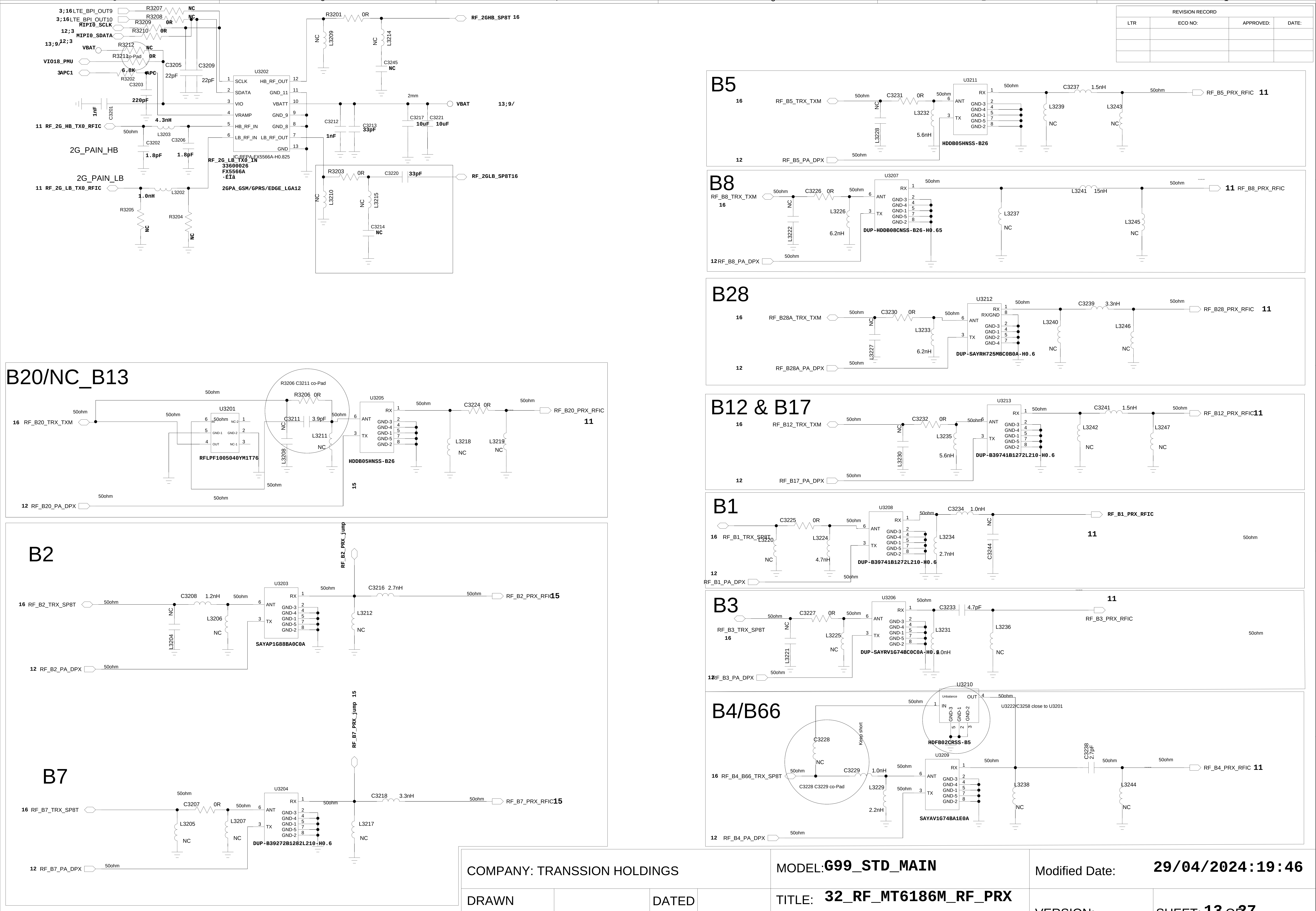
RF_MT6186M_RF_TX

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS		MODEL: G99_STD_MAIN	Modified Date: 28/04/2024:21:43
DRAWN	DATED	TITLE: 31_RF_MT6186M_RF_TRX	VERSION:
CHECKED	DATED < >	Confidentiality CONFIDENTIAL	SHEET: 12 OF 27

swap B5&8 @MMMB PA 20220119



COMPANY: TRANSSION HOLDINGS			MODEL: G99_STD_MAIN	Modified Date: 29/04/2024:19:46
DRAWN		DATED	TITLE: 32_RF_MT6186M_RF_PRX	VERSION:
CHECKED		DATED < >	Confidentiality CONFIDENTIAL	SHEET: 13 OF 27

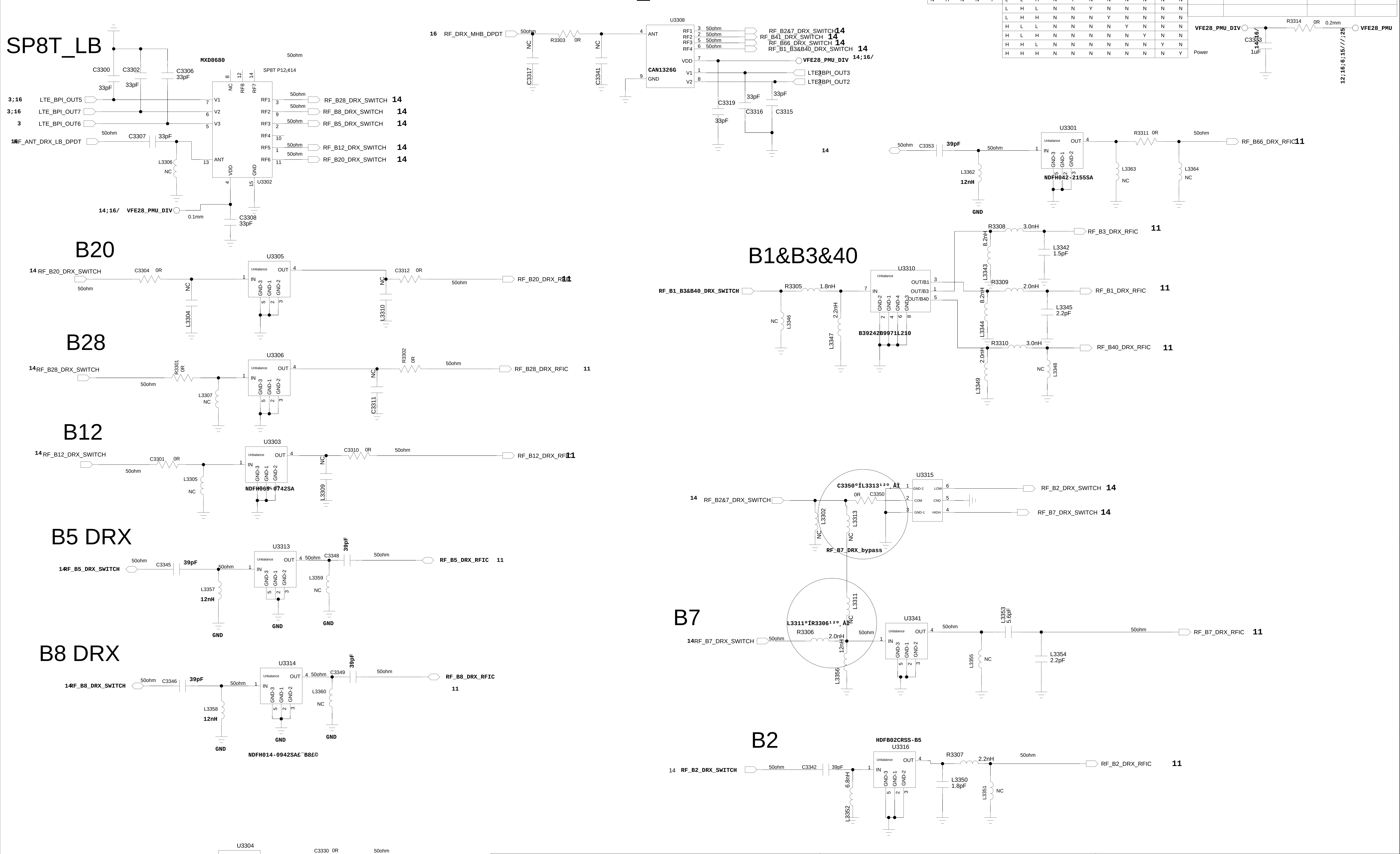
RF_MT6186M_RF_DRX

SP8T_MHB

VC1613 control logic				
VC1	VC2	RF1	RF2	RF3
H	L	Y	N	N
H	H	N	Y	N
N	H	N	N	Y

SP8T control logic										
SP6T control logic										
V1	V2	V3	RF1	RF2	RF3	RF4	RF5	RF6	RF7	RF8
L	L	L	Y	N	N	N	N	N	N	N
L	L	H	N	Y	N	N	N	N	N	N
L	H	L	N	N	Y	N	N	N	N	N
L	H	H	N	N	N	Y	N	N	N	N
H	L	L	N	N	N	N	Y	N	N	N
H	L	H	N	N	N	N	N	Y	N	N
H	H	L	N	N	N	N	N	N	Y	N
H	H	H	N	N	N	N	N	N	N	Y

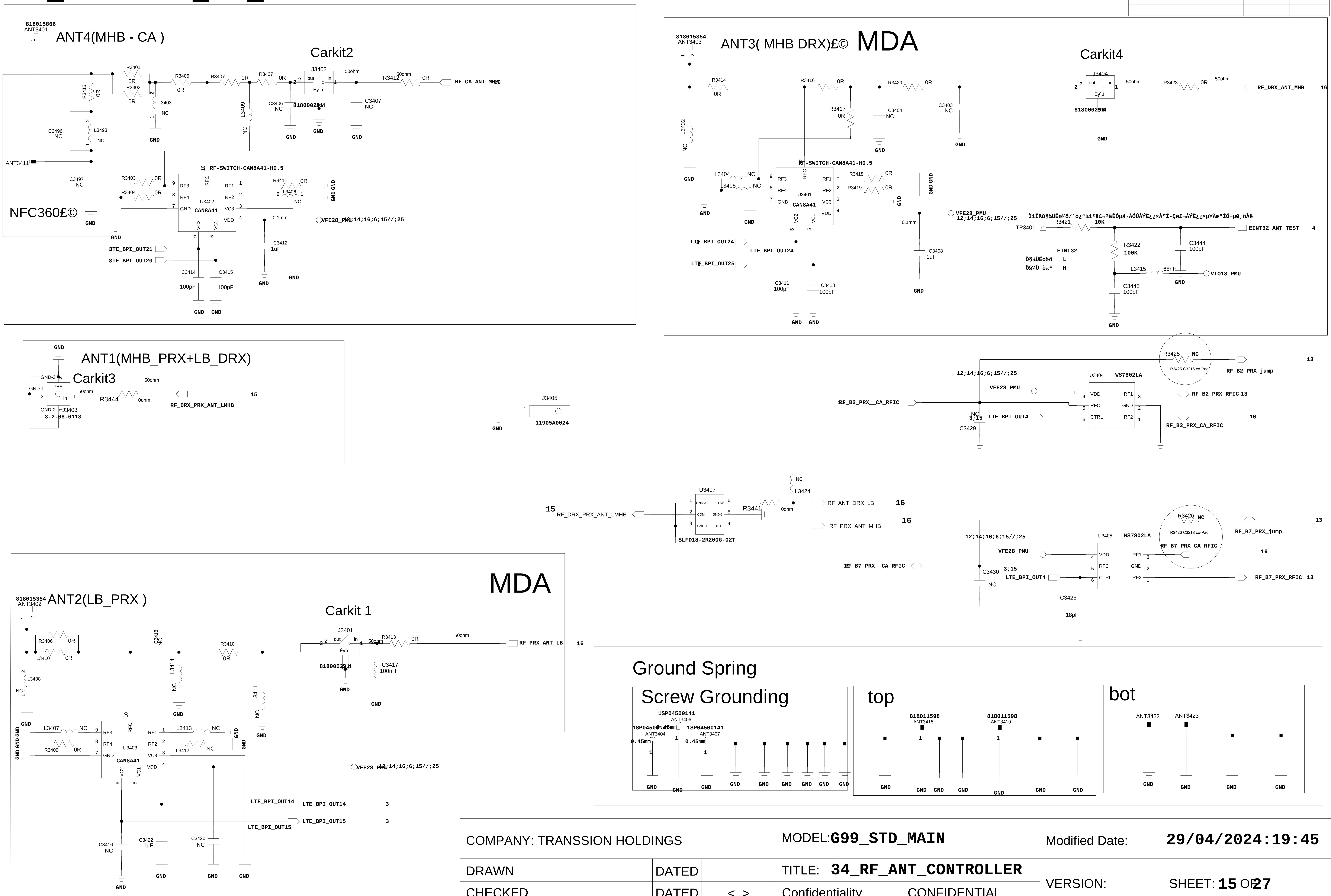
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS		MODEL: **G99_STD_MAIN**	Modified Date: **29/04/2024:10:13**
DRAWN	DATED	TITLE: **33_RF_MT6186M_RF_DRX**	VERSION:
CHECKED	DATED < >	Confidentiality CONFIDENTIAL	SHEET: **14** OF **27**

RF_MT6186M_RF_ANT MDA

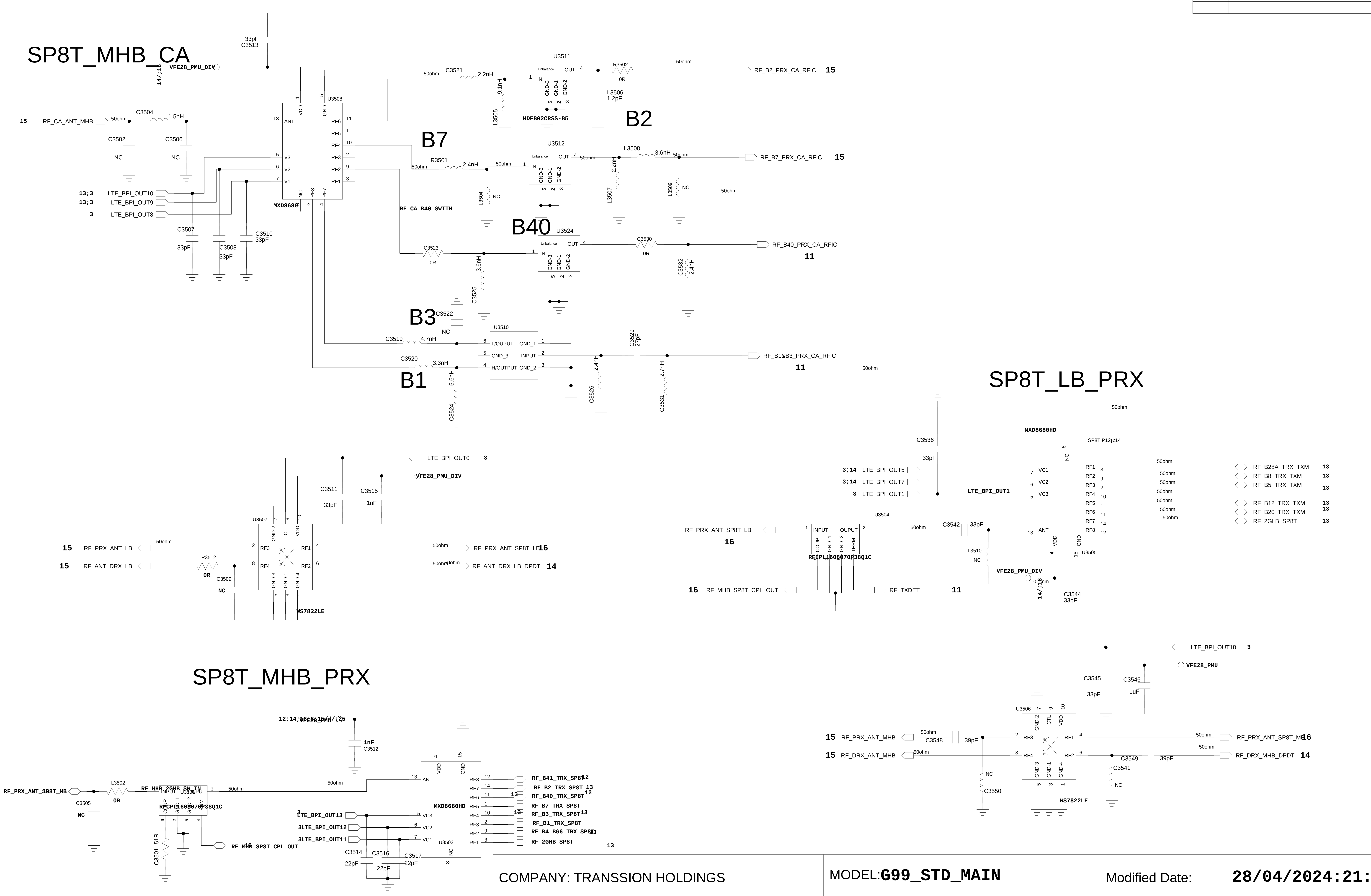
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS			MODEL: G99_STD_MAIN		Modified Date: 29/04/2024:19:45
DRAWN		DATED	TITLE: 34_RF_ANT_CONTROLLER		VERSION:
CHECKED		DATED < >	Confidentiality	CONFIDENTIAL	SHEET: 15 OF 27

RF_MT6186M_RF_MHB_CA

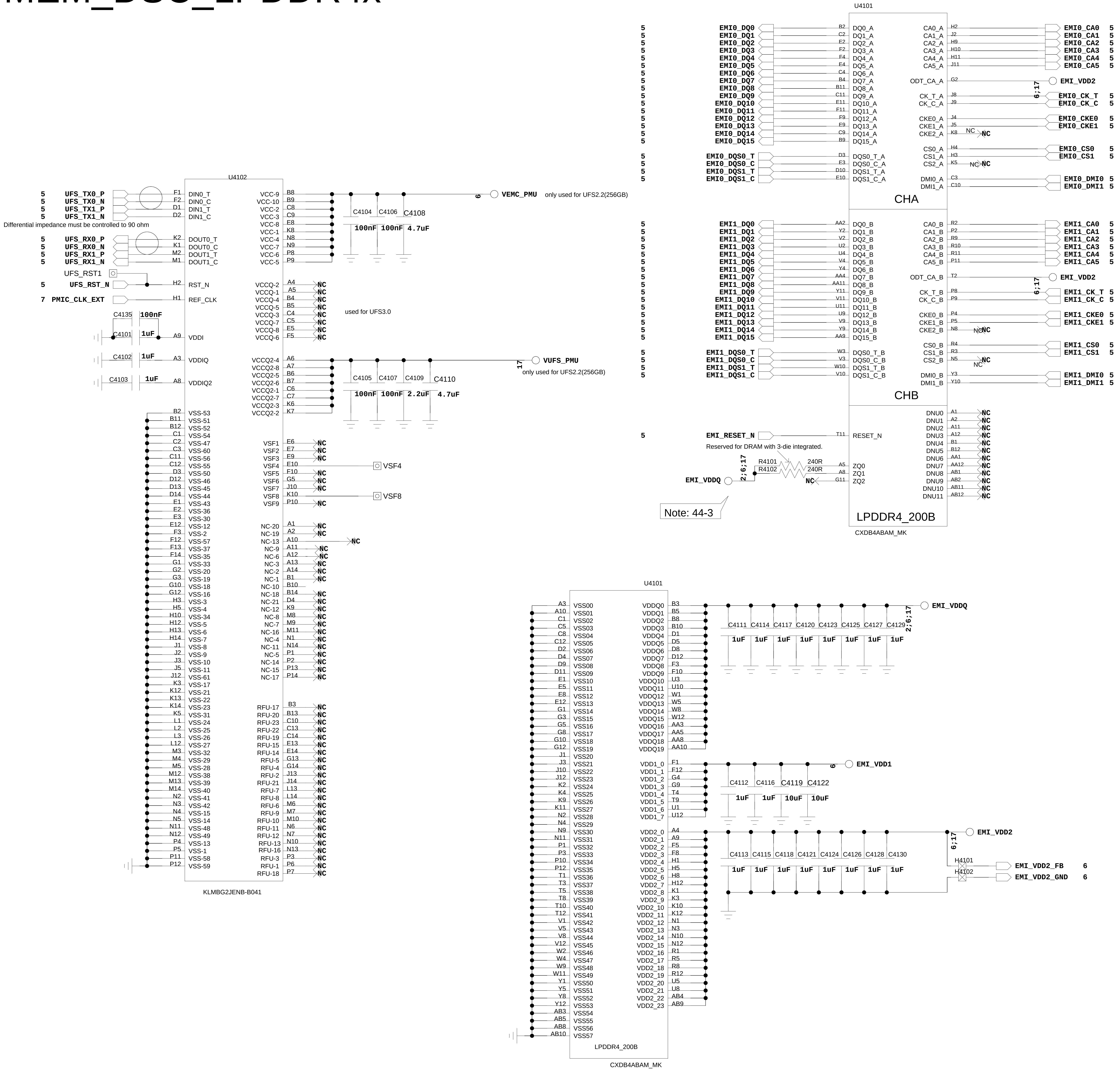
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



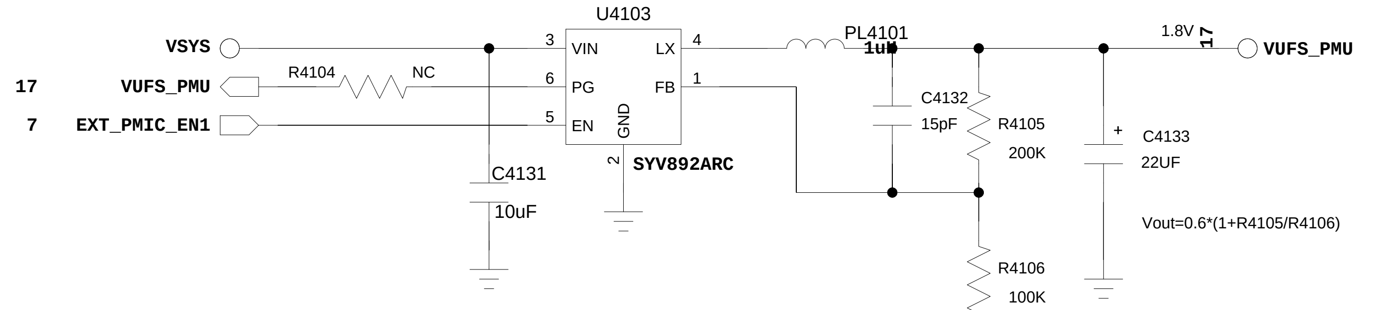
COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN	Modified Date: 28/04/2024:21:42
DRAWN		DATED		TITLE: 35_RF_MT6186M_RF_MHB_CA	
CHECKED		DATED	< >	Confidentiality CONFIDENTIAL	VERSION:	SHEET: 16 OF 27

MEM_DSC_LPDDR4x

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



For UFS 2.2 VCCQ2

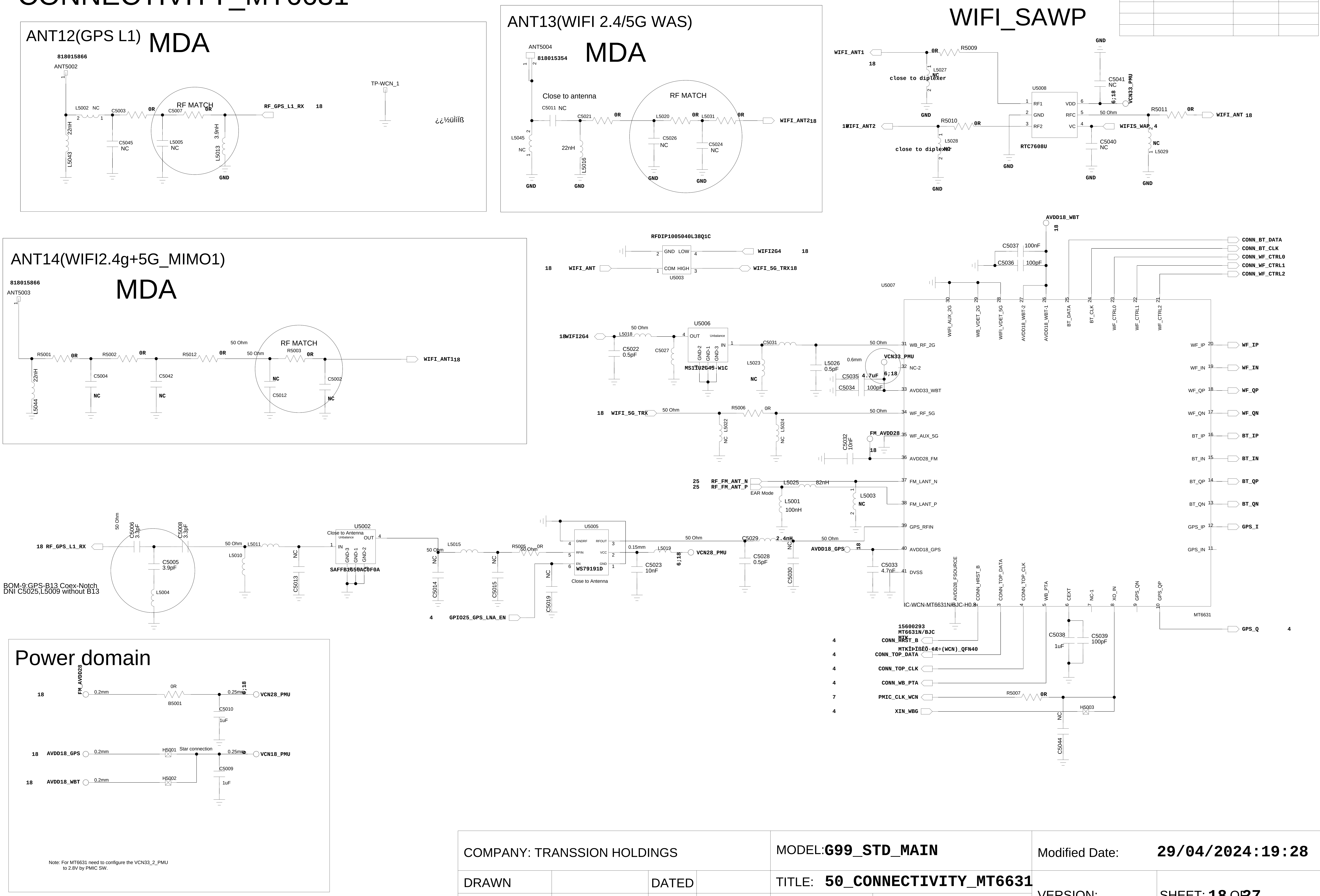


	R4104
DIO6905	0R
DIO6905B	NC
WD10721	NC
SYV892	NC

COMPANY: TRANSSION HOLDINGS	MODEL: G99_STD_MAIN	Modified Date:	02/05/2024:14:38
DRAWN	DATED	TITLE: 41_MEMORY_DSC_LPDDR4X	
CHECKED	DATED < >	Confidentiality CONFIDENTIAL	VERSION: SHEET: 17 OF 27

CONNECTIVITY_MT6631

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



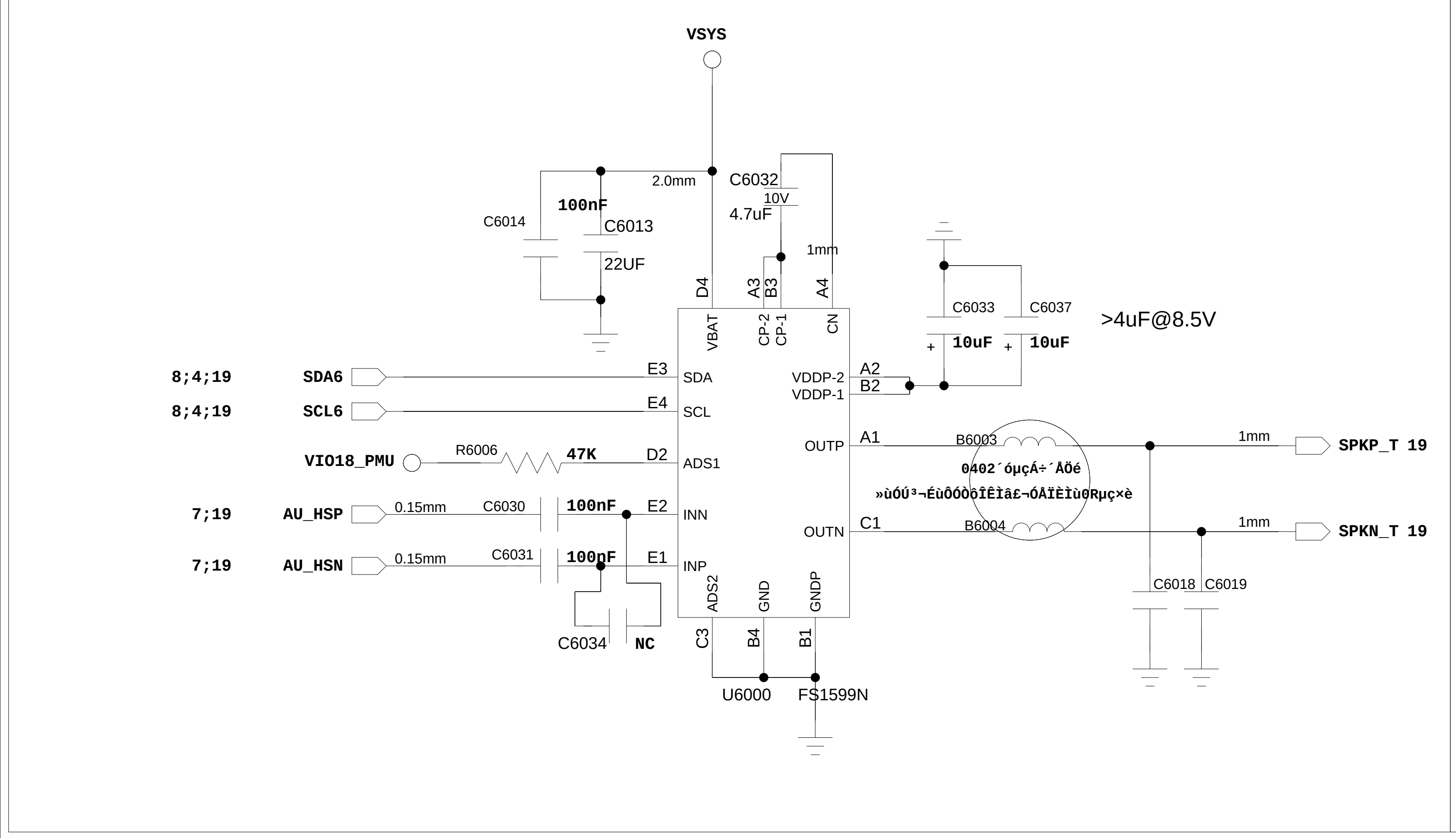
COMPANY: TRANSSION HOLDINGS			MODEL: G99_STD_MAIN	Modified Date: 29/04/2024:19:28
DRAWN		DATED	TITLE: 50_CONNECTIVITY_MT6631	VERSION:
CHECKED		DATED < >	Confidentiality CONFIDENTIAL	SHEET: 18 OF 27

PERI_AUDIO_IO

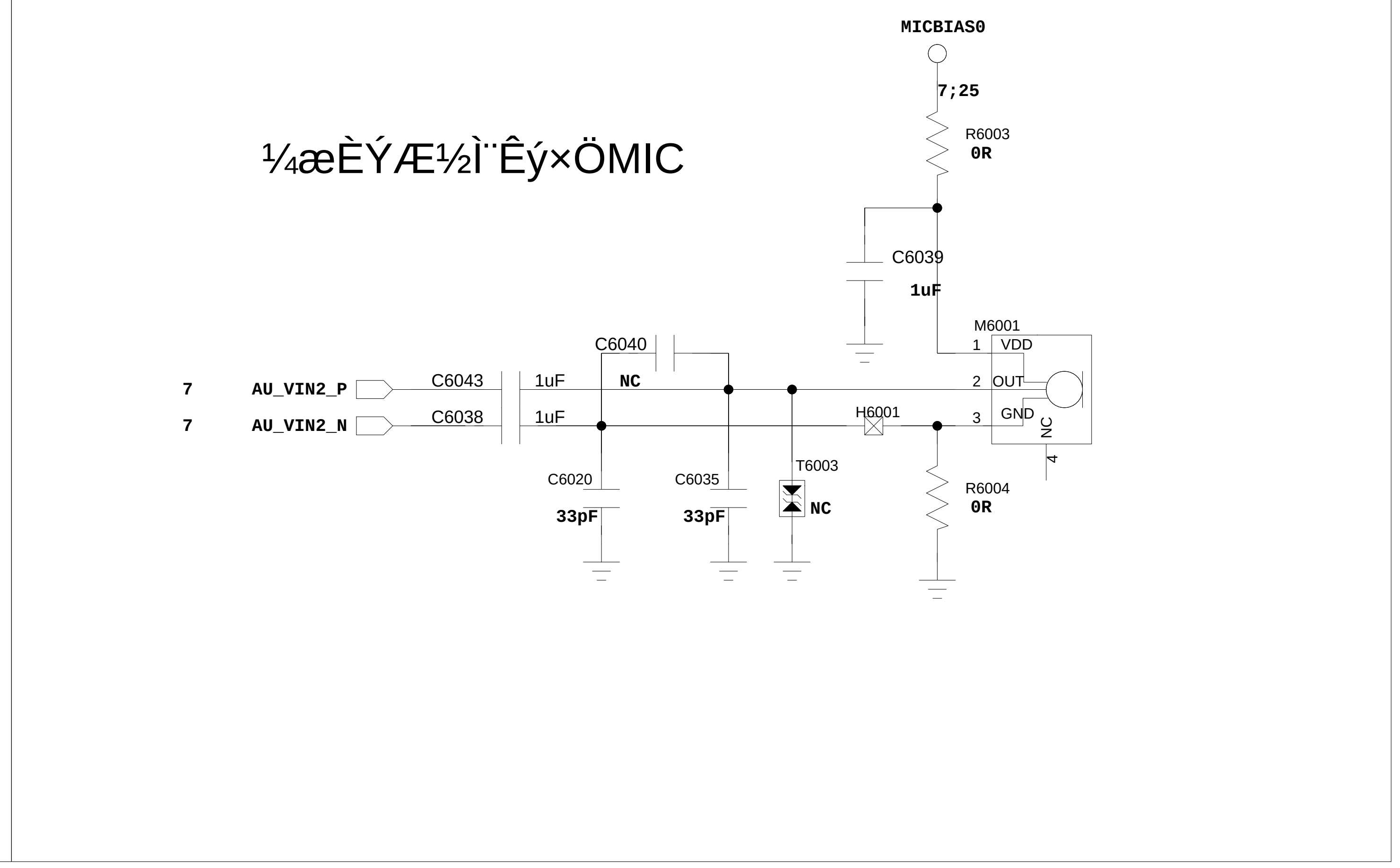
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

Audio PA
UP SPK

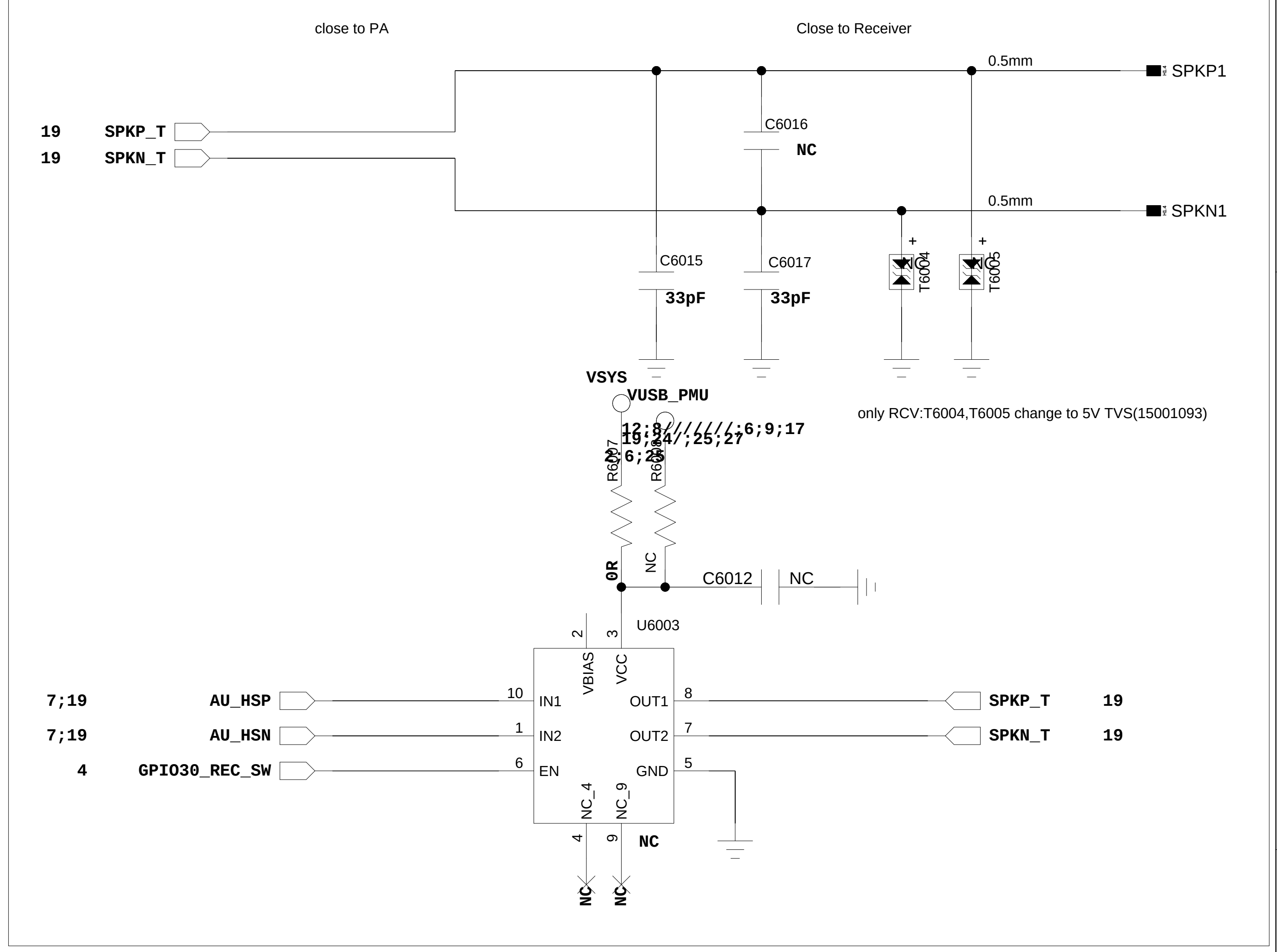
FS1599£¨I2C ADDR 01101A2A1



SUB MIC-ACC

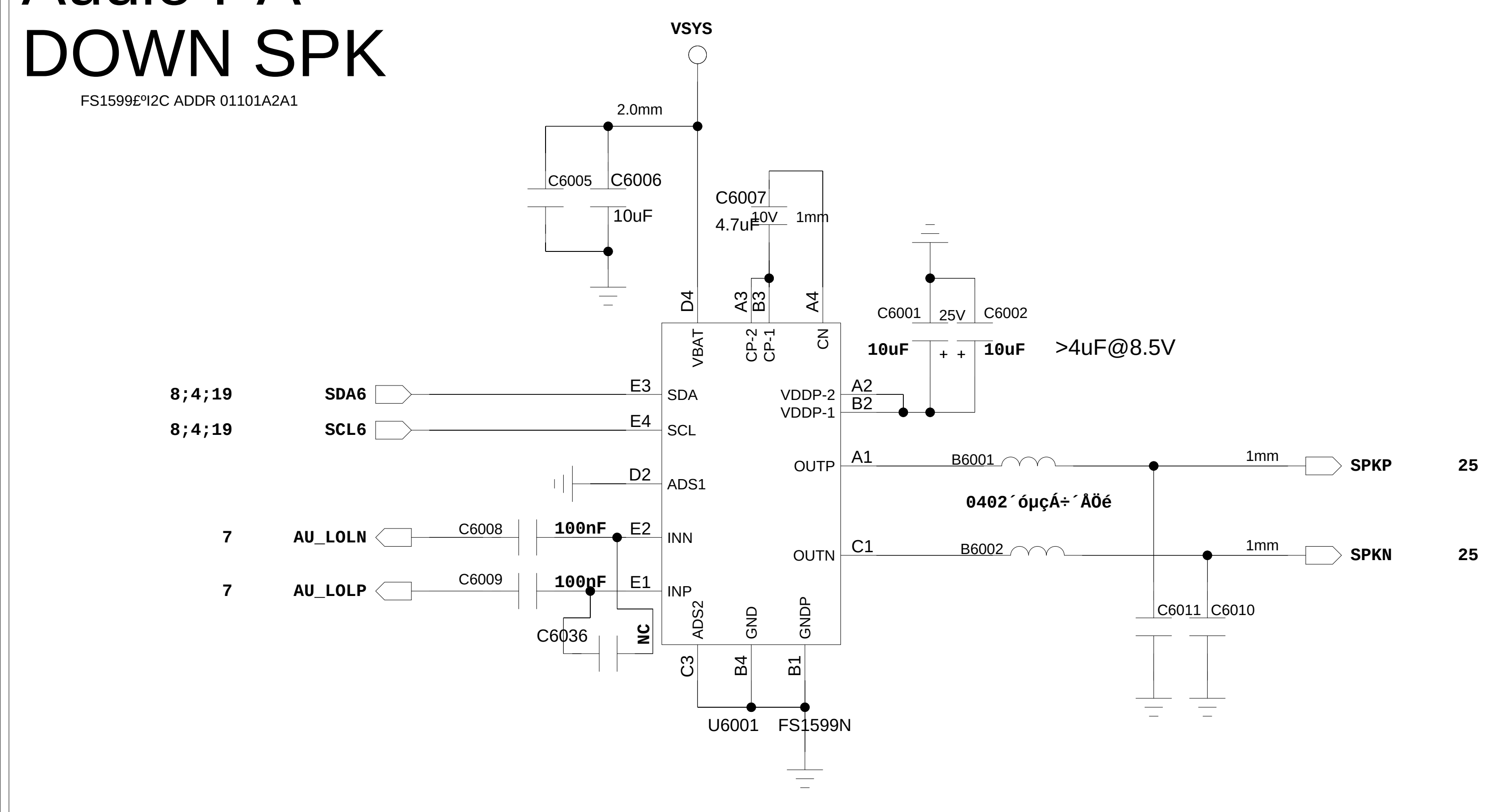


REC&SPK 2in1

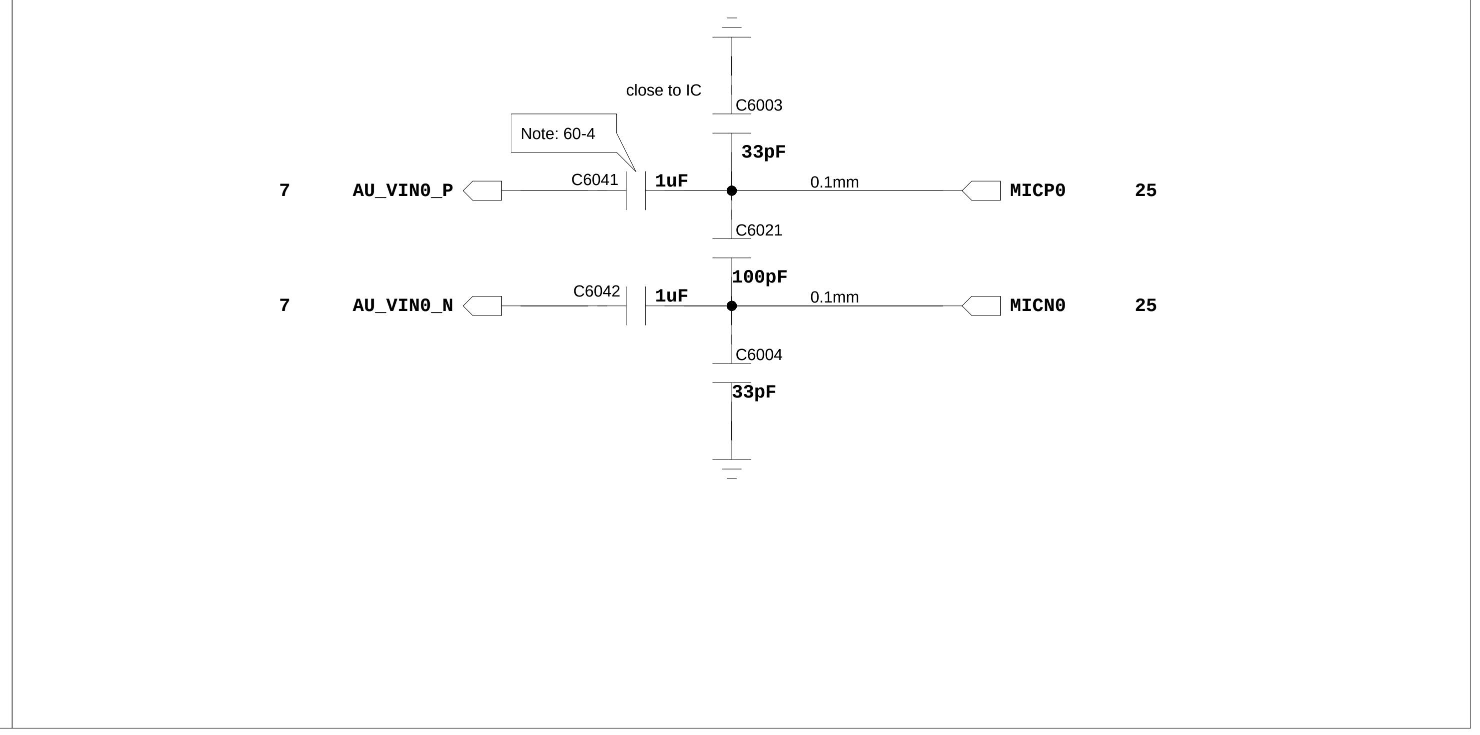


Audio PA
DOWN SPK

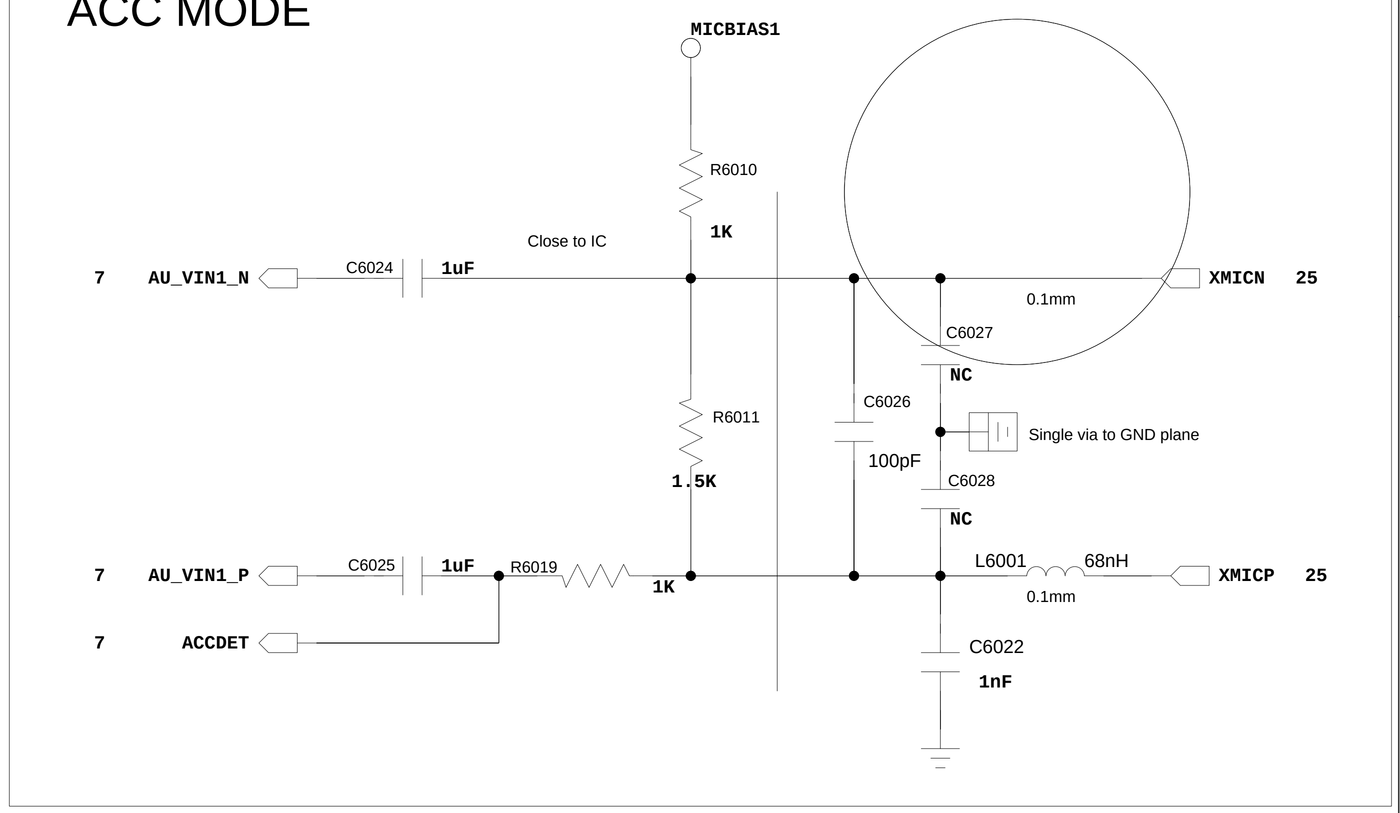
FS1599£¨I2C ADDR 01101A2A1



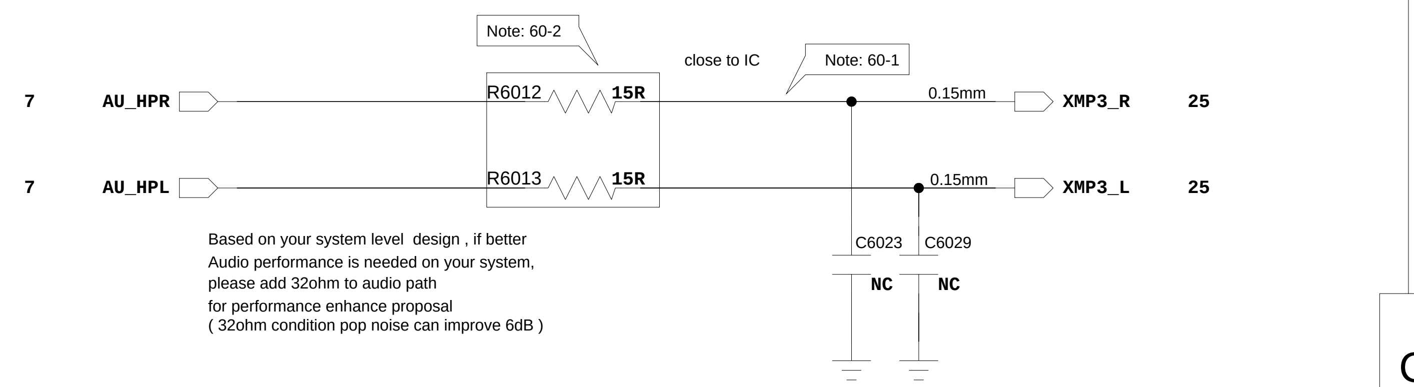
MAIN MIC



Earphone MICPHONE
ACC MODE



Earphone Receiver



Based on your system level design , if better
Audio performance is needed on your system,
please add 32ohm to audio path
for performance enhance proposal
(32ohm condition pop noise can improve 6dB)

Schematic design notice of "60_PERI_AUDIO_IO" page.	
Note 60-1:	B6009 B6010 B6013 B6014 needs change to "BLM18BD102SN1" for high THD performance(-90dB), but this BOM change will result in FM RSSI 10dB degraded.
Note 60-2:	To reserve a resistor in HPL and HPR in series connection both in order to optimize headphone pop noise. The recommended value of this resistor is 33R.
Note 60-3:	Layout trace from MT6353 ball J3 AUDREFN to Audio jack GND must surround shield with GND.
Note 60-4:	0.1/1uF for ACC mode(1uF for WB_AMR Speech/0.1uF for NB_AMR Speech),0R for DCC mode.

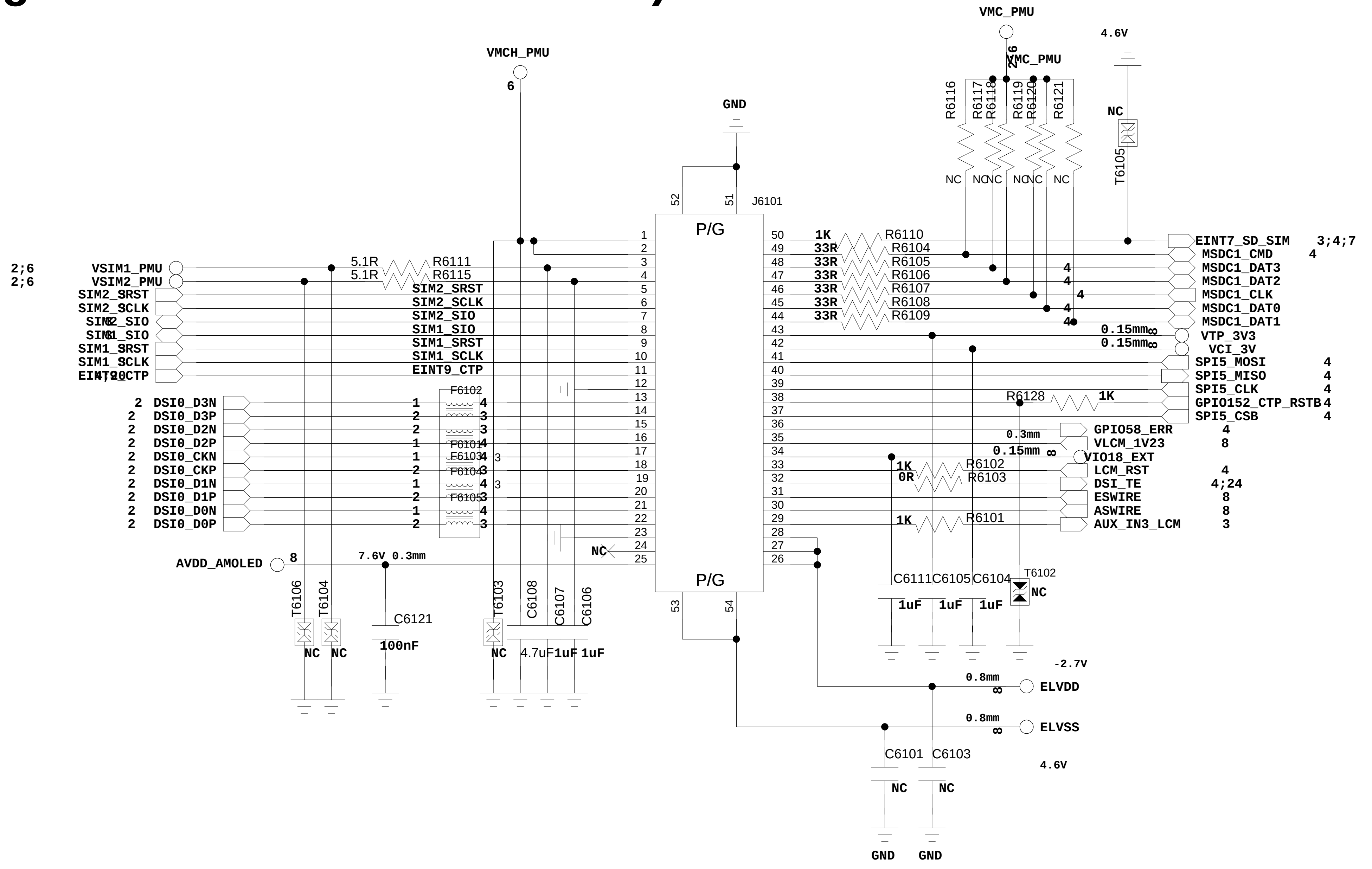
COMPANY: TRANSSION HOLDINGS				MODEL:**G99_STD_MAIN**	Modified Date: **17/04/2024:14:51**
DRAWN		DATED		TITLE: **60_PERI_AUDIO**	VERSION:
CHECKED		DATED	< >	Confidentiality CONFIDENTIAL	SHEET: **19** OF **27**

PERI_LCM_CTP_FP

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

LCD_CTP_INTERFACE

Exchange IO2£¨LCM/CTP/SIM/TF)



PUSH OUT	PUSH IN	SD DETECT
L	H	EINT

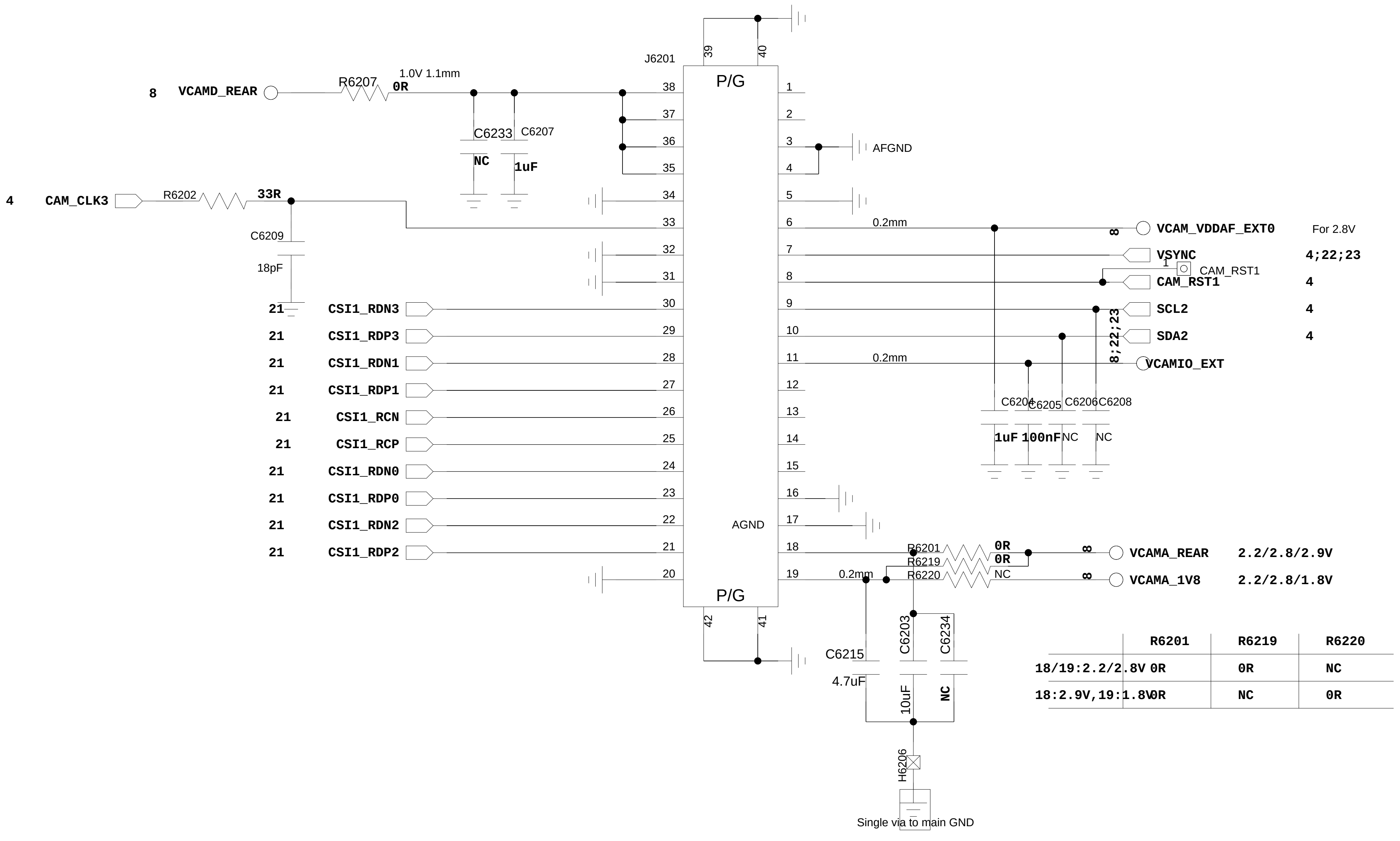
FINGER PRINT

COMPANY: TRANSSION HOLDINGS			MODEL: **G99_STD_MAIN**	Modified Date: **17/04/2024:15:12**
DRAWN		DATED	TITLE: **61_PERI_LCM_CTP_FP**	VERSION:
CHECKED		DATED < >	Confidentiality CONFIDENTIAL	SHEET: **20** OF **27**

PERI_CAMERA_I

(CAM I 108M AF)

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



	R6201	R6219	R6220
18/19:2.2/2.8V	0R	0R	NC
18:2.9V,19:1.8V	0R	NC	0R

		AVDD	DVDD	IOVDD	AFVDD	OIS-VDD	I2C addr
50M 17201892	S5KJN1SQ03-FGX9 AW86016CSR TD24C128-C2LPTR	2.8V67.7mA	1.05V760.7mA	1.8V4mA	2.8V103mA		Sensor:0x20(W)/0x21(R) VCM:0x18(W)/0x19(R) EEPROM:0xA0(W)/0xA1(R)
50M 17201891	Hi-5022Q DW9800S GT24P128F-2CLI-TR	2.8V68mA	1.1V243mA	1.8V22mA	2.8V105mA		Sensor:0x44(W)/0x45(R) VCM:0x18(W)/0x19(R) EEPROM:0xA0(W)/0xA1(R)
50M 17201906	S5KJN1SQ03-FGX9 AW86016CSR P24C128F-D4H-MIR	2.8V67.7mA	1.05V760.7mA	1.8V4mA	2.8V105mA		Sensor:0x20(W)/0x21(R) VCM:0x18(W)/0x19(R) EEPROM:0xA0(W)/0xA1(R)
108M	S5KHM6SX03-FGX9 AW8601CSR GT24P128F-2CSLI-TR	2.2V120mA	1.0V1100mA	1.8V38mA	2.8V103mA		Sensor:0x20(W)/0x21(R) VCM:0x18(W)/0x19(R) EEPROM:0xA0(W)/0xA1(R)

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COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN	Modified Date: 29/04/2024:20:39
DRAWN		DATED		TITLE: 62_PERI_CAMERA_I	VERSION:
CHECKED		DATED	< >	Confidentiality CONFIDENTIAL	SHEET: 21 OF 27

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

CAM II (8M)

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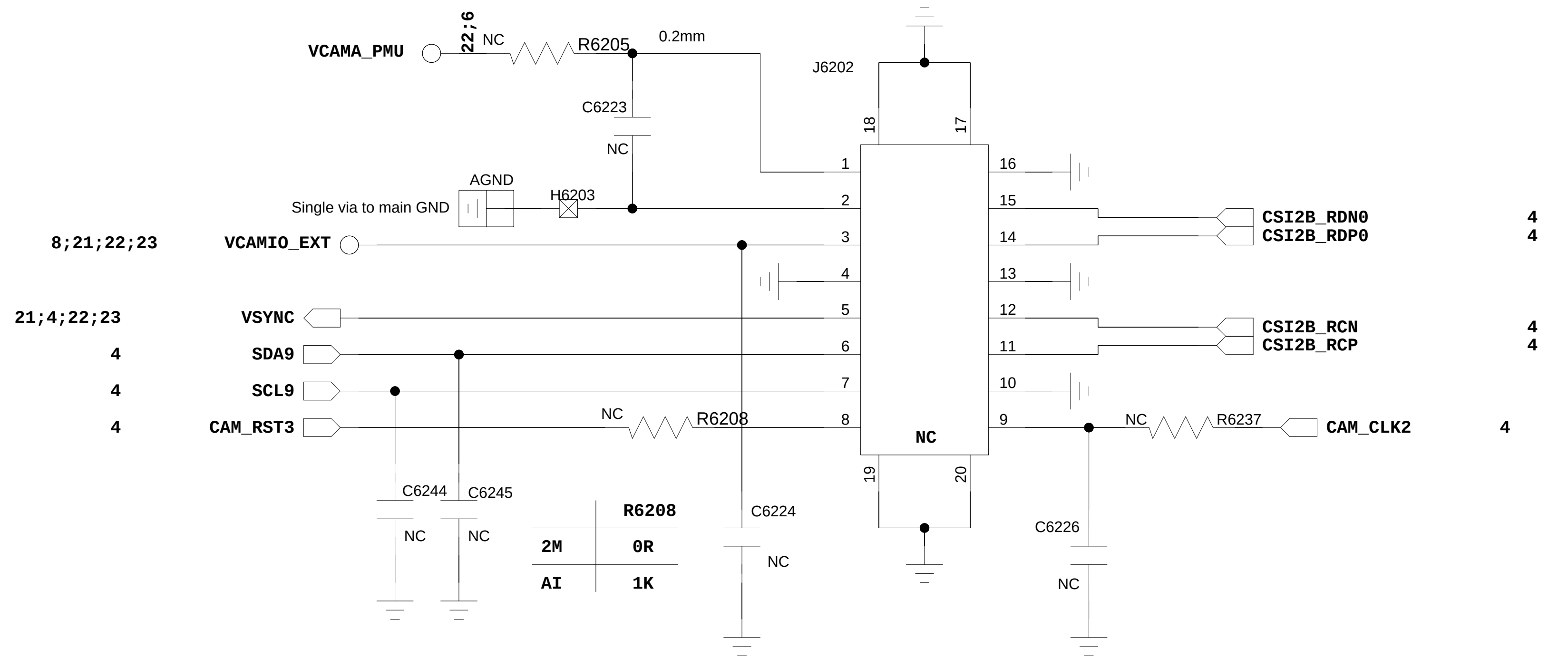
CAM III (2M/AI)

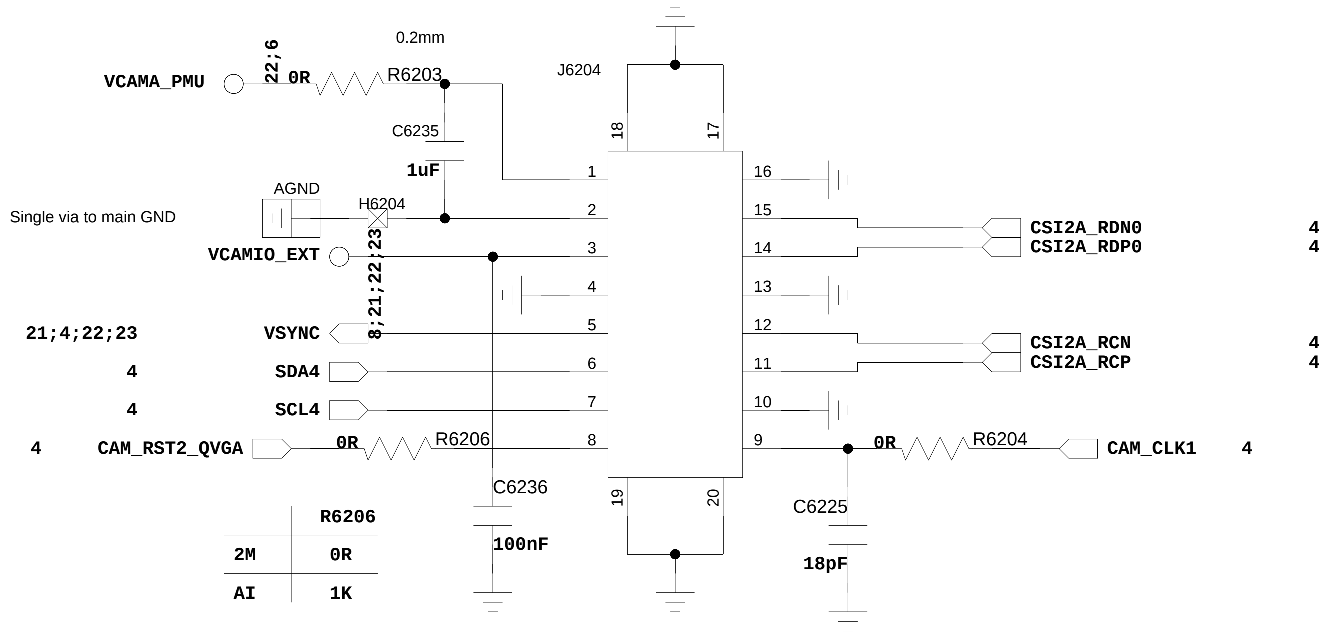
17201873
17201874
17201889



		AVDD	DVDD	IOVDD	I2C addr
2MÐé»¯	2M(17201866) GC02M3-C24Y0	2.8V,40mA	/	1.8V,70mA	sensor:0x20(W)/0x21(R)
	2M(17201867) GC02M3-C24YA	2.8V,40mA	/	1.8V,70mA	sensor:0x20(W)/0x21(R)
2MÎ¢¾à	2M(17201868) GC02M3-C24YA	2.8V,40mA	/	1.8V,70mA	sensor:0x78(W)/0x79(R)

CAM IV (2M/AI)

17201873
17201874
17201889

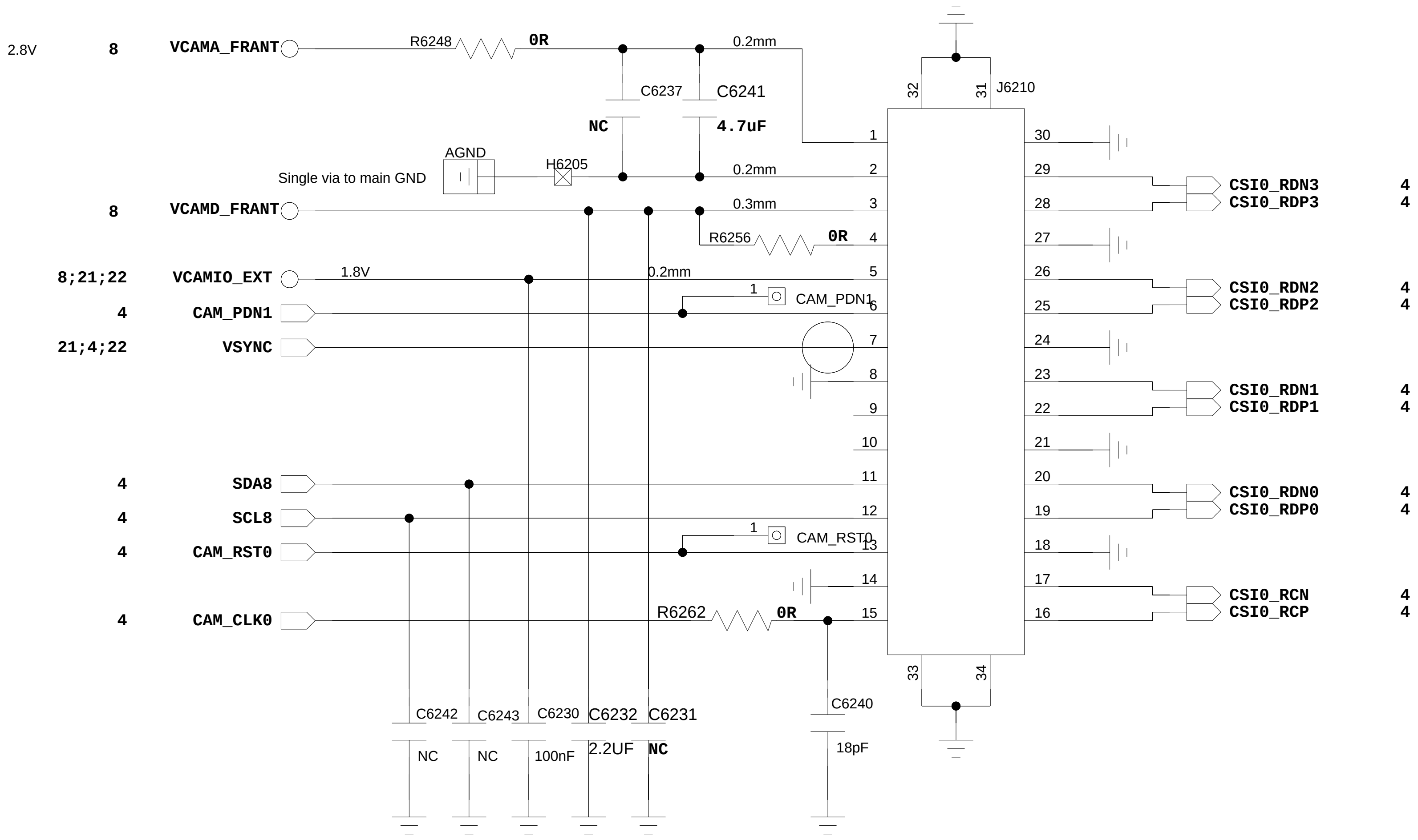


COMPANY: TRANSSION HOLDINGS		MODEL: G99_STD_MAIN		Modified Date:	17/04/2024:14:51
DRAWN		DATED		TITLE: 62_PERI_CAMERA_II		VERSION:	SHEET: 22 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

PERI_CAMERA_III

FRONT MAIN CAMERA (8M/32M)



108M/50M+2M/Q+2M/Q

		AVDD	DVDD	IOVDD	AF	4Pin	I2C addr
8M	17201890 GC08A8-MADD0	2.8V40mA	1.2V 110mA	1.8V2mA		NC	Sensor:0x24(W)/0x25(R)
8M	17201895 OV08D10-GA5A-001A	2.8V50mA	1.2V 90mA	1.8V1mA		NC	Sensor:0x6C(W)/0x6D(R) EEPROM:0xA0(W)/0xA1(R)
8M	1720xxxx SC820CS-CR7NN01 GT24P64E-2CSLI-TR	2.8V30mA	1.2V 120mA	1.8V1mA		NC	Sensor:0x6C(W)/0x6D(R) EEPROM:0xA0(W)/0xA1(R)
32M	17201856 GC32E1-WA1XA P24C64F-A4H-MIR	2.8V55mA	1.2V120mA	1.8V5mA		DVDD	Sensor:0x94(W)/0x95(R) EEPROM:0XA2(W)/0XA3(R)
32M	17201895 S5KKD1SP03-FGX9 P24064F-A4H-MR	2.8V	1.05V	1.8V		DVDD	Sensor:0xAC(W)/0xAD(R) EEPROM:0XA2(W)/0XA3(R)
32M	1720xxxx GC32E2-WA12A P24C64F-A4H-MIR	2.8V	1.1V	1.8V		DVDD	Sensor:0x94(W)/0x95(R) EEPROM:0XA2(W)/0XA3(R)
13M	1720xxxx GC13A0-WA1NA GT24P64E-2CSLI-TR	2.8V 45mA	1.2V 115mA	1.8V 5mA		DVDD	Sensor:0x72(W)/0x73(R) EEPROM:0XA2(W)/0XA3(R)

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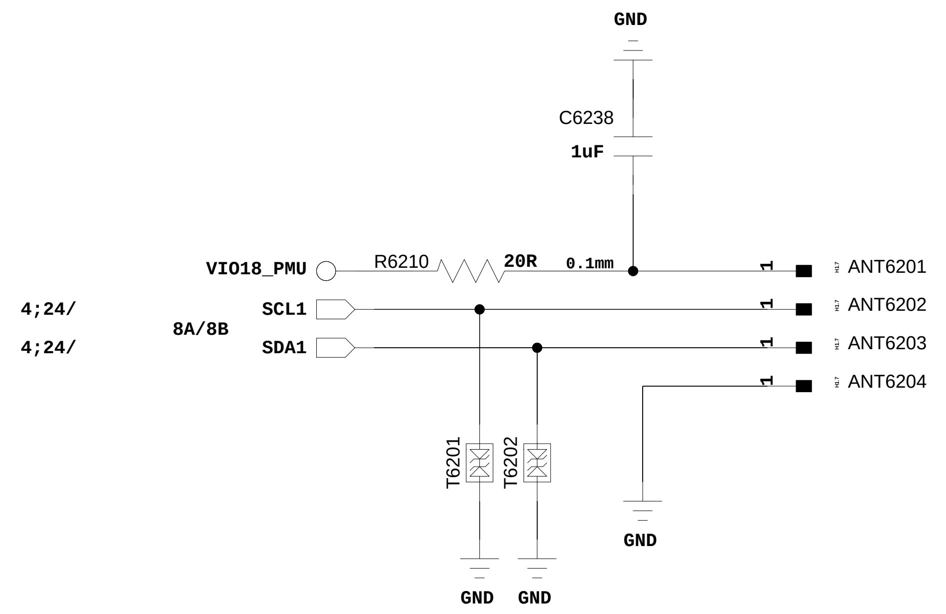
STK31860



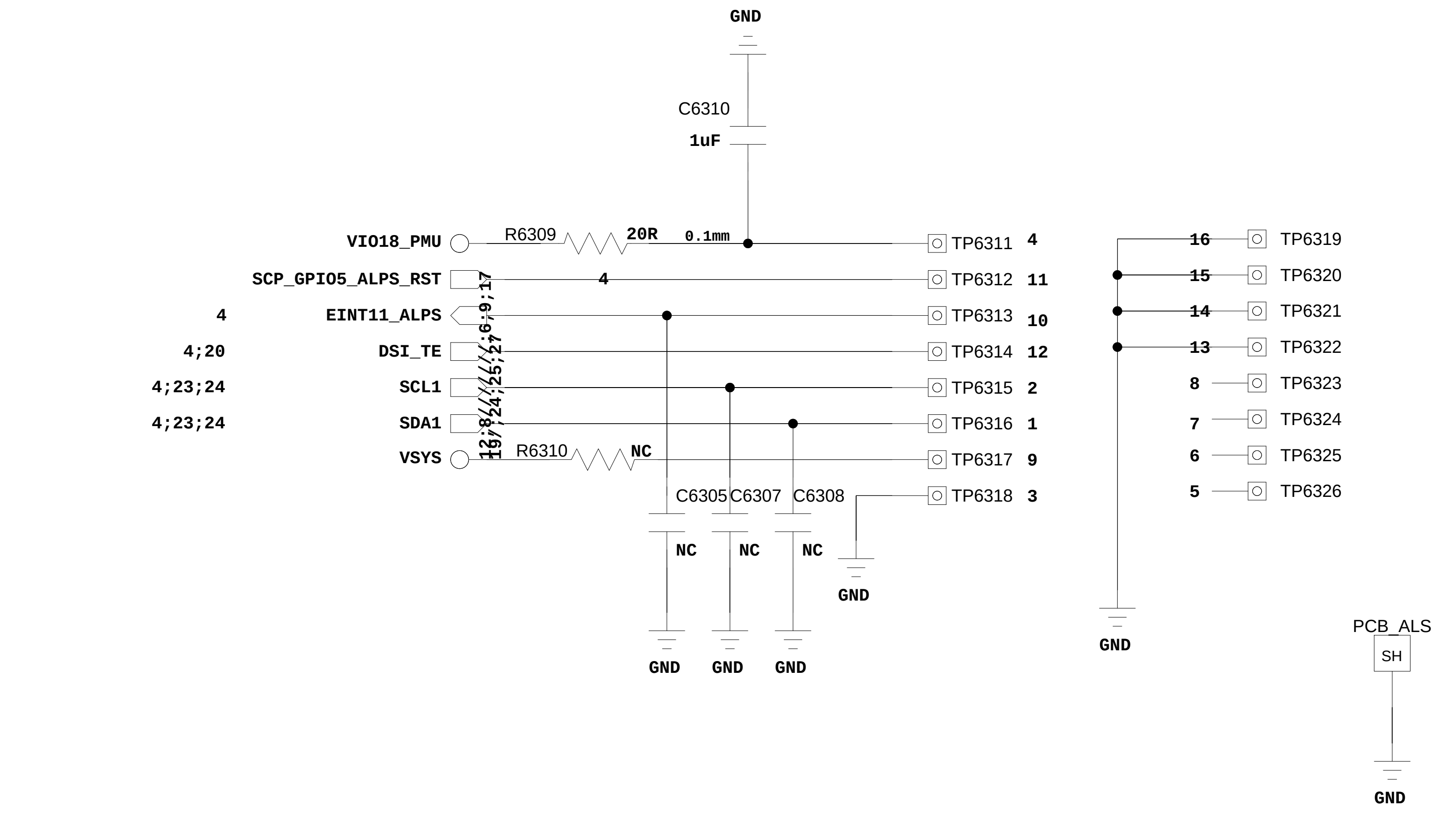
COMPANY: TRANSSION HOLDINGS		MODEL: G99_STD_MAIN	Modified Date: 17/04/2024:14:51
DRAWN	DATED	TITLE: 62_PERI_CAMERA_III	VERSION:
CHECKED	DATED < >	Confidentiality CONFIDENTIAL	SHEET: 23 OF 27

PERI_SENSORS

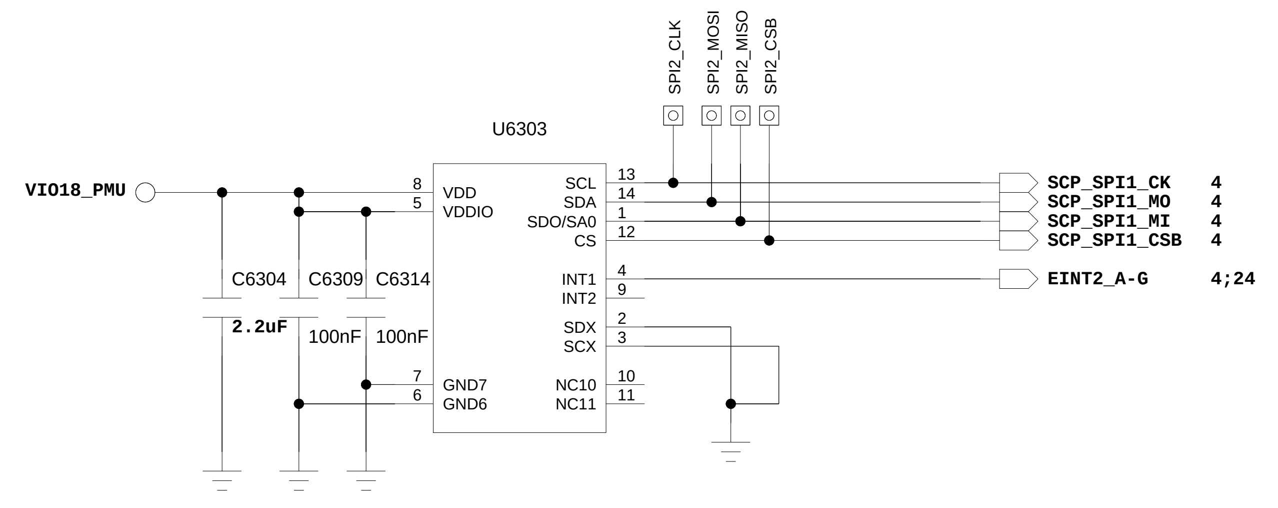
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

AL& PS Sensor

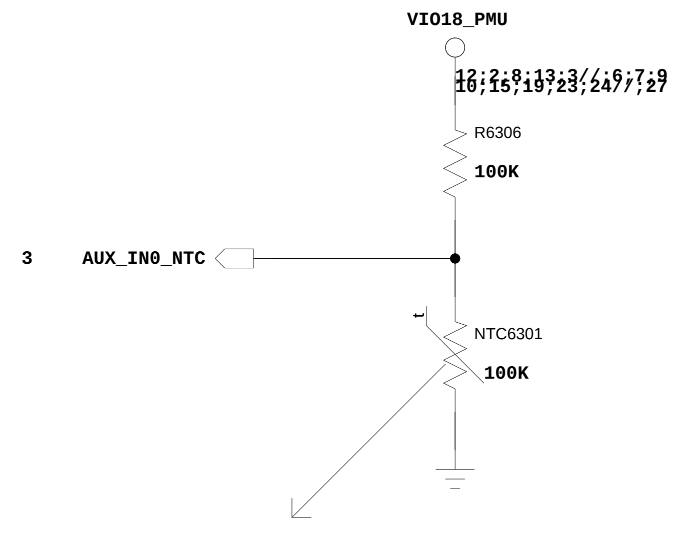
STK35F62:I2C address: Write:0x90, Read:0x91



G-Sensor + Gyro Sensor



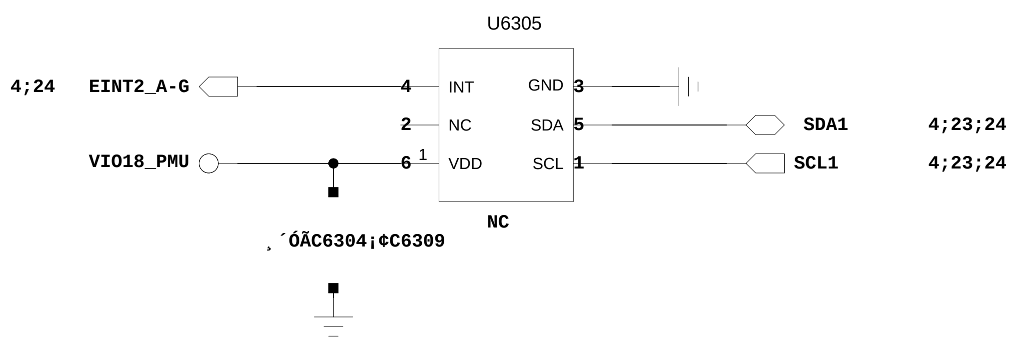
Thermistor to sense AP temperature



1. NTC6301must keep a distance about 6-8 mm away from BB and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

G-Sensor

MXC4005XC:I2C ADDRESS:0x2A(Write)/0x2B(Read)



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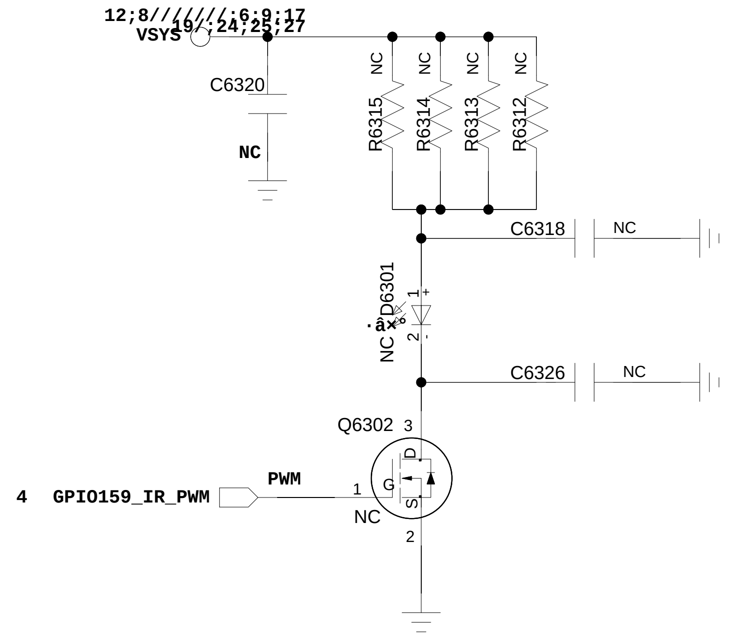
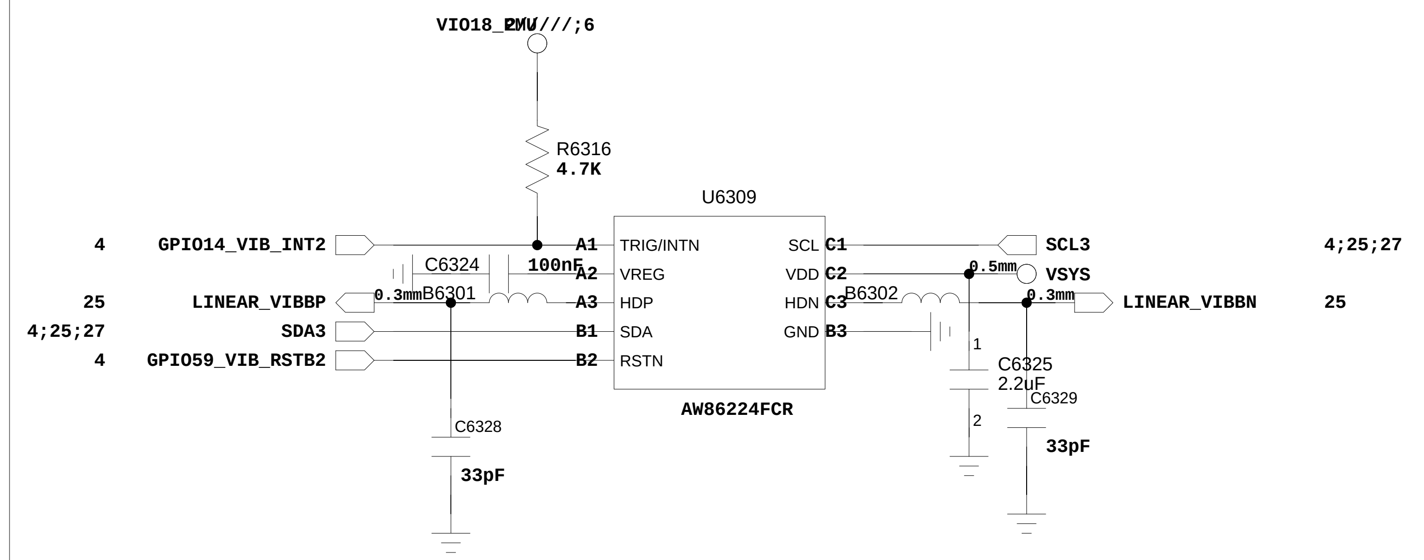


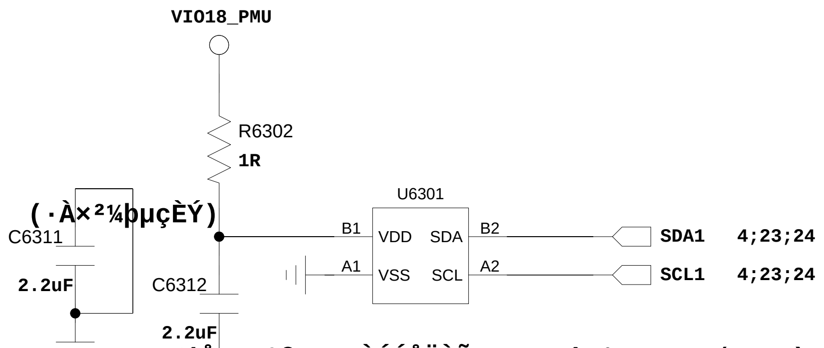
VIB Z Linear Motor Driver

AW86224FCR:I2C ADDRESS:0xB0(Write)/0xB1(Read)



M-Sensor(COMPASS)

MMC5603:I2C ADDRESS:0x60(Write)/0x61(Read)
QMC6308:I2C ADDRESS:0x58(Write)/0x59(Read)



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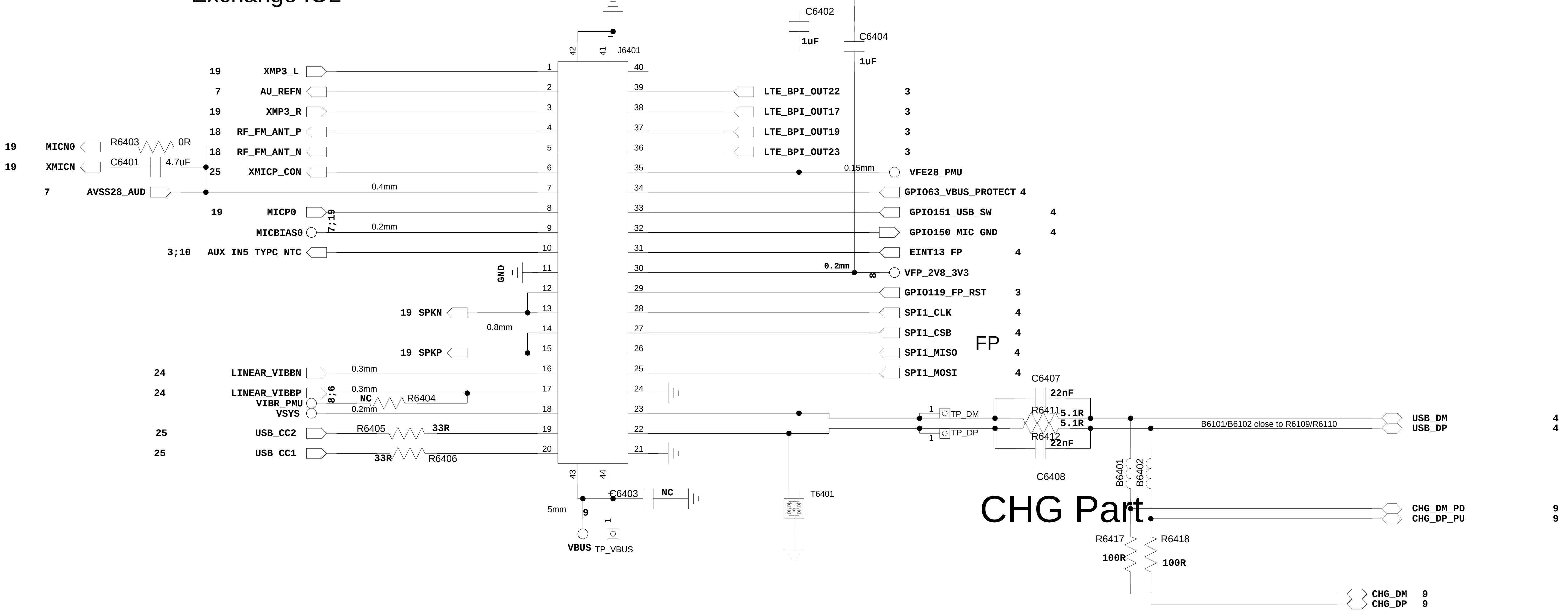
COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN	Modified Date: 25/04/2024:17:00
DRAWN		DATED		TITLE: 63_PERI_SENSORS	VERSION:
CHECKED		DATED	< >	Confidentiality CONFIDENTIAL	SHEET: 24 OF 27

PERI_EXCH_IO

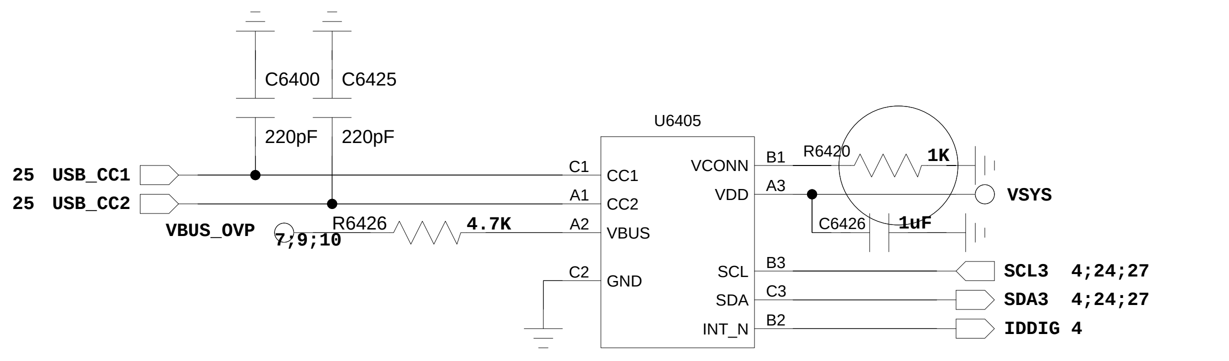
Exchange IO2



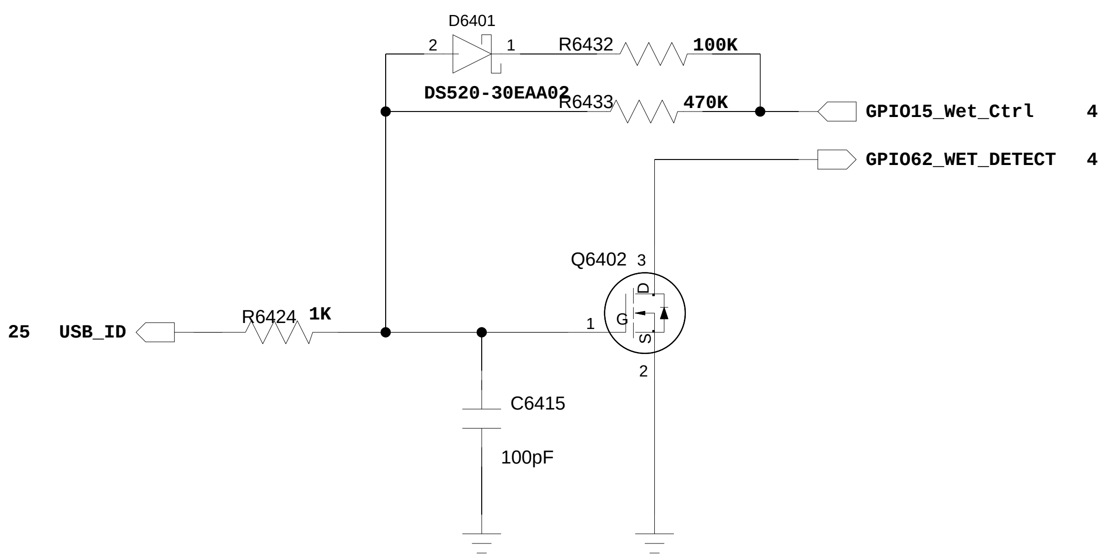
CC Logic

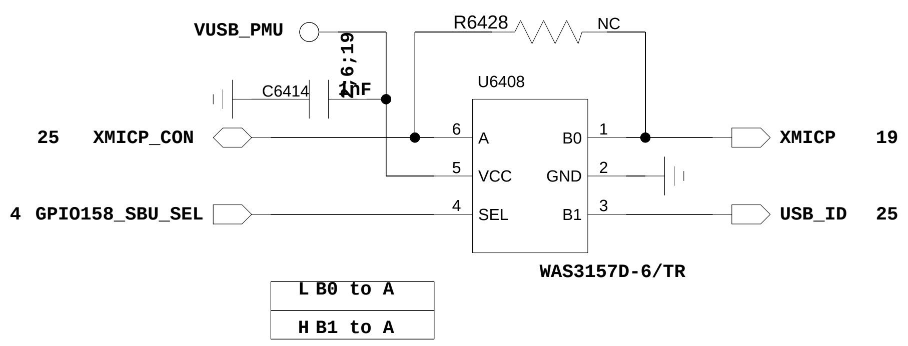
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

RT1715:I2C ADDRESS:0x9C(Write)/0x9D(Read)



Wet Detect





COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN	Modified Date: 17/04/2024:14:51
DRAWN		DATED		TITLE: 64_PERI_EXCH_IO	VERSION:
CHECKED		DATED	< >	Confidentiality: CONFIDENTIAL	SHEET: 25 OF 27

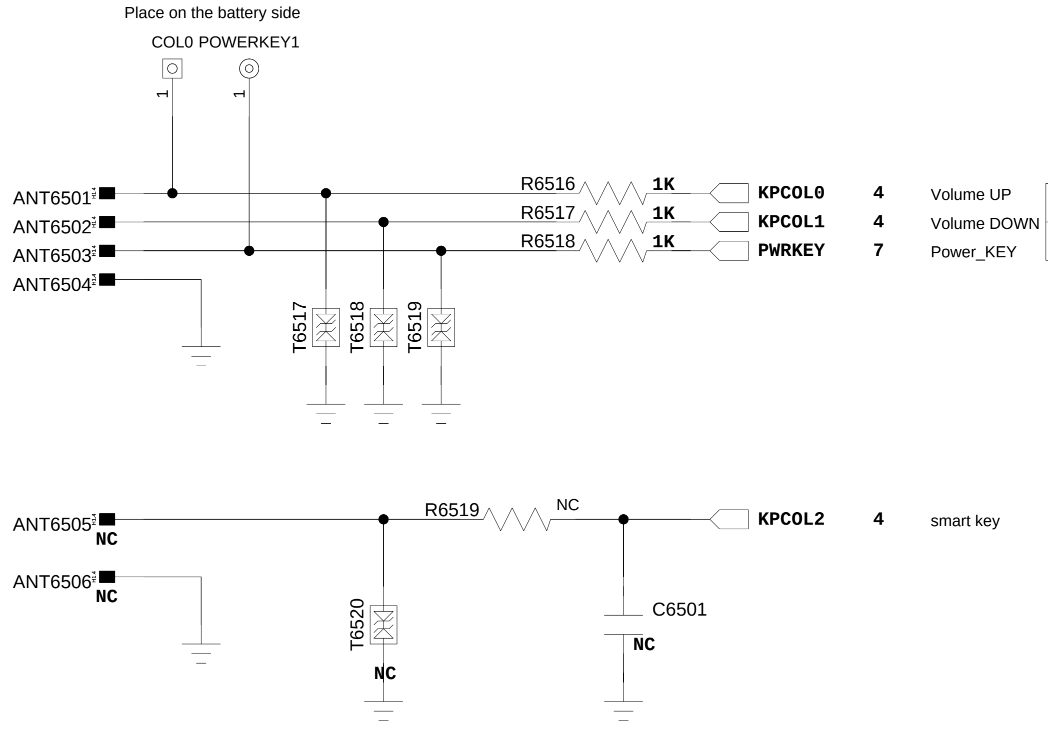
PERI_SIM_SD_KEYPAD

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

SIM1/2-SD-CARD

SD DETECT	PUSH IN	PUSH OUT
EINT	H	L

SIDEKEY



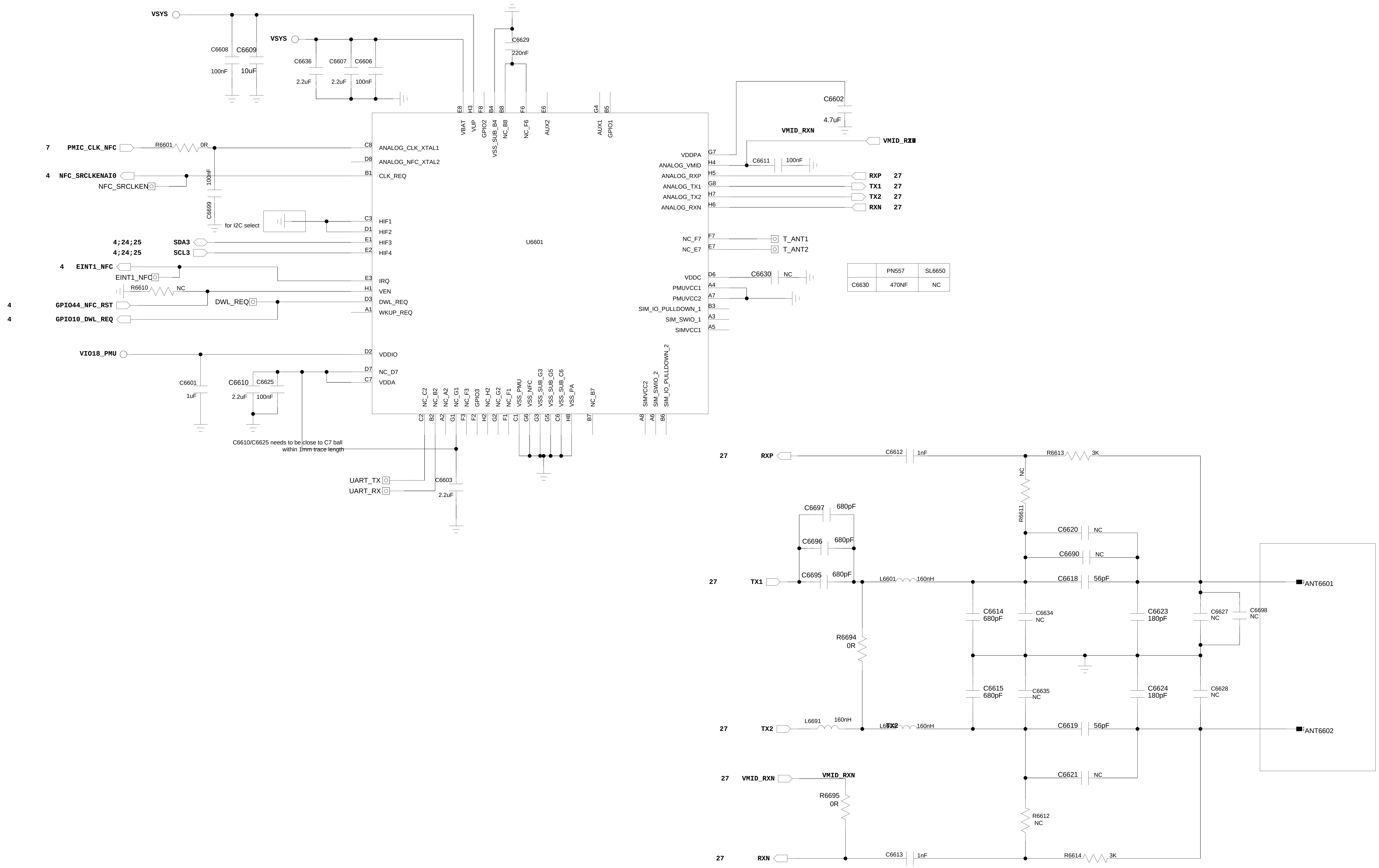
	COL0	COL1
GND	UP	DOWN

COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN	Modified Date: 17/04/2024:14:51
DRAWN		DATED		TITLE: 65_SD_SIM_KEY	VERSION:
CHECKED		DATED	< >	Confidentiality: CONFIDENTIAL	SHEET: 26 OF 27

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

PERI_NFC

THN31FGB1N:I2C address: Write:0X50, Read:0X51



	PN557	SL6650
C6630	470NF	NC

COMPANY: TRANSSION HOLDINGS			MODEL: G99_STD_MAIN		Modified Date: 03/06/2024:09:48
DRAWN		DATED	TITLE: 66_NFC		VERSION:
CHECKED		DATED < >	Confidentiality	CONFIDENTIAL	SHEET: 27 OF 27